

Applied Statistics Problem Sets



Exercise 1.1

1. The following table gives the heights (in inches) of fathers (X) and their son's (Y).

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

Calculate the correlation coefficient between them.

2. A computer operator while calculating correlation coefficient between two variables X and Y from 30 pairs of observations obtained the following results-

$$n = 30, \sum X = 95, \sum Y = 115, \sum X^2 = 650, \sum Y^2 = 675, \sum XY = 705$$

It was however, later discovered at the time of checking that he had copied down two pairs as (6, 14) and (9, 8), while the correct values were (8, 12) and (8, 7). Obtain the incorrect and the correct correlation coefficients.

3. Let $-6x + 4y - 2 = 0$ and $4x - 5y - 7 = 0$ be the two lines of regression. Then find-
- the mean values of X and Y ,
 - the correlation coefficient,
 - the angle between the lines,
 - y for $x = -5$ and x for $y = 7$.
4. The industrial productions and demands for 7 industries in a year are given as below-

Productions (in crore tons)	55	56	58	59	60	60	62
Demands (in crore tons)	35	38	38	39	44	43	45

Then find-

- the correlation coefficient.
 - the regression coefficients.
 - the two regression lines.
 - the production corresponding to demand 40 and the demand corresponding to the production 57.
5. If $4x - 5y + 33 = 0$ and $20x - 9y - 107 = 0$ are two lines of regression, find-
- the mean values of x and y .
 - the regression coefficients b_{xy} and b_{yx} .
 - the correlation coefficient.
 - the angle between the two lines.
 - the standard deviation of y if the variance of x is 9.
 - the value of x for $y = 4$.
6. The following table gives the heights (in inches) of fathers (X) and their son's (Y).

X	65	66	67	67	68	69	70	72
Y	67	68	65	68	72	72	69	71

Find-

- the two regression coefficients.
- the two regression lines.
- the angle between the two lines.
- the height of the father for the height of the son 75 inches.

Exercise 2.1

1. Find the regression equation of X_2 on X_1 and X_3 , given the following results-

Trait	Mean	SD	r_{12}	r_{23}	r_{31}
X_1	28.02	4.42	0.80	—	—
X_2	4.91	1.10	—	-0.56	—
X_3	594	85	—	—	-0.40

where X_1 = seed per acre, X_2 = Rainfall in inches, X_3 = accumulated temperature above $42^{\circ}F$.

2. A household appliance manufacturer wants to analyze the relationship between total sales and the company's two primary means of advertising television and magazines from the following data-

X	67	72	63	65	71	64
Y	22	25	29	31	29	32
Z	33	40	25	30	35	29

where X_1 = Total sales, X_2 = expenditure on television, X_3 = expenditure on radio. All amounts are in lakhs. Calculate the total sales corresponding to the expenditures $X_2 = 34, X_3 = 20$.

3. The electric power $X(Kw)$ consumed each month by a chemical plant is thought to be related to the ambient temperature $Y(^{\circ}C)$ and the number of working days Z in the month. The past year's historical data are available and are presented in the following table-

X	87	93	95	88	94	98
Y	72	78	83	70	79	85
Z	25	28	23	29	26	30

Find the number of working days corresponding to the consumption $X = 87Kw$ and the ambient temperature $Y = 77^{\circ}C$.

4. In a study of a random sample of 120 students, the following results are obtained-

$$\begin{aligned}\bar{X}_1 &= 68, & \bar{X}_2 &= 70 & \bar{X}_3 &= 74 \\ \sigma_1^2 &= 100 & \sigma_2^2 &= 25 & \sigma_3^2 &= 81 \\ r_{12} &= 0.60 & r_{13} &= 0.70 & r_{23} &= 0.65\end{aligned}$$

If X_1, X_2, X_3 denote the percentage of marks obtained by a student in the test I , test II and in the final examination respectively, then find the percentage of the marks of a student in the final exam if he gets 60% and 67% marks in I and II tests respectively.

5. The electric power X consumed each month by a chemical plant is thought to be related to the ambient temperature Y and the number of working days Z in the month. The past year's historical data are available and are presented in the following table-

$$\begin{aligned}\bar{X} &= 92, & \bar{Y} &= 68 & \bar{Z} &= 24 \\ \sigma_x^2 &= 16 & \sigma_y^2 &= 9 & \sigma_z^2 &= 4 \\ r_{xy} &= 0.60 & r_{xz} &= 0.70 & r_{yz} &= 0.65\end{aligned}$$

Predict the power consumption for a month in which the ambient temperature is 75 and the number of working days is 18.

6. An engineer at a semiconductor company wants to model the relationship between the three parameters: Emitter- $RS(X_1)$, Base- $RS(X_2)$ and Emitter-to-Base $RS(X_3)$. The data are shown in the table-

Parameter	Mean	σ	r_{12}	r_{13}	r_{23}
X_1	6.2	1.5	0.45	—	—
X_2	18.4	4.1	—	0.28	—
X_3	27.5	5.3	—	—	0.54

Predict Emitter-to-Base $RS(X_3)$, when Emitter- $RS(X_1) = 5.8$, Base- $RS(X_2) = 15.3$.

7. An experiment was conducted to determine if the final weight X of an animal can be predicted after a given period of time on the basis of the initial weight Y of the animal and the amount of feed Z that was eaten. The following data, measured in kilograms, were recorded:

$$\begin{aligned}\sigma_X &= 14.98, \sigma_Y = 4.33, \sigma_Z = 43.69, \\ r_{12} &= 0.78, r_{23} = 0.60, r_{31} = 0.88 \\ \bar{X} &= 82.5, \bar{Y} = 37.9, \bar{Z} = 241.7\end{aligned}$$

Estimate the initial weight of the animal if the final weight and the amount of feed are 75 kg and 187 kg respectively.

8. Estimate the rainfall (inches) corresponding to the seeds per acre 12 kg and the temperature $38^\circ C$ from the following results-

Trait	Mean	SD	r_{12}	r_{23}	r_{31}
X_1	28.02 gm	4.42	0.80	—	—
X_2	4.91 inch	1.10	—	-0.56	—
X_3	$36^\circ C$	85	—	—	-0.40

where X_1 =seeds (kg) per acre, X_2 =Rainfall (inches), X_3 =temperature ($^\circ C$).

9. If X_1 (liters) is the consumption of milk per head, X_2 (Rs.) the mean price of milk and X_3 (00's) the per capita income. The standard deviations and the total correlation coefficients are calculated as-

$$\begin{aligned}\sigma_1 &= 7.22, \sigma_2 = 5.47, \sigma_3 = 6.87, \\ r_{12} &= -0.83, r_{23} = 0.92, r_{31} = -0.61, \\ \bar{X}_1 &= 8, \bar{X}_2 = 36, \bar{X}_3 = 62.\end{aligned}$$

Calculate the regression equation of X_2 on X_1 and X_3 and estimate the mean price of the milk corresponding to the consumption $X_1 = 17$ liters and per capita income $X_3 = 72$ Rs.

10. From the heights X_1 , weights X_2 and ages X_3 of a group of students, the following standard deviations and correlation coefficients were obtained-

$$\begin{aligned}\sigma_1 &= 2.8 \text{ inches}, \sigma_2 = 12 \text{ lbs}, \sigma_3 = 1.5 \text{ years}, \\ r_{12} &= 0.75, r_{23} = 0.54, r_{31} = 0.43 \\ \bar{X}_1 &= 5.8 \text{ inches}, \bar{X}_2 = 42 \text{ lbs}, \bar{X}_3 = 17 \text{ years}\end{aligned}$$

Find the regression equation of X_3 on X_1 and X_2 and estimate the value of X_3 for $X_1 = 5.6$ and $X_2 = 55$.

11. A household appliance manufacturer wants to analyze the relationship between total sales and the company's two primary means of advertising television and magazines. They obtained the following results-

Trait	Mean	SD	r_{12}	r_{23}	r_{31}
X_1	67	7.5	0.75	—	—
X_2	28	5.3	—	-0.28	—
X_3	32	4.9	—	—	0.64

where X_1 = Total sales, X_2 = expenditure on television, X_3 = expenditure on radio. All amounts are in lakhs. Calculate the total sales corresponding to the expenditures $X_2 = 34, X_3 = 20$.

12. In a tri-variate distribution of 18 observations, the following data was obtained; $r_{12} = 0.77, r_{13} = 0.72, r_{23} = 0.52$. Find the partial correlation coefficients.
13. If $r_{12} = 0.80, r_{13} = -0.40$ and $r_{23} = -0.56$, find the values of all the partial and multiple correlation coefficients.
14. In a trivariate distribution involving X_1, X_2 and X_3 , if $r_{12} = 0.57, r_{13} = -0.28, r_{23} = -0.63, \sigma_1 = 23, \sigma_2 = 14$ and $\sigma_3 = 7$ then find-
- all the multiple correlation coefficients,
 - all the partial correlation coefficients.
15. The simple correlation coefficients between temperature X_1 , corn yield X_2 and rainfall X_3 are $r_{12} = 0.59, r_{13} = 0.46, r_{23} = 0.77$. Calculate $r_{23.1}$ and $R_{1.32}$.

Exercise 3.1

1. Find the probability distribution of the random variable X denoting the numbers of heads in a random experiment of tossing three coins together.
2. Find the probability distribution of the random variable X denoting the sum of the numbers on the upper faces in random experiment of throwing two dice together.
3. Find the probability distribution of the total number of heads obtained in four tosses of a balanced coin.
4. Let X denotes the number of red balls when five balls are drawn at random with replacement from a bag containing 8 red balls and 6 green balls. Find the probability distribution of X .
5. Determine whether the following function can serve as the probability distribution of a random variable X –
 - (i) $f(x) = \frac{x^2}{30}, x = 0, 1, 2, 3, 4$
 - (ii) $f(x) = \frac{x-2}{5}, x = 1, 2, 3, 4, 5$
 - (iii) $f(x) = \frac{2x}{k(k+1)}, x = 1, 2, 3, \dots, k$
6. Find the probability mass function of the total number of heads obtained in four tosses of a balanced coin.
7. Find the probability mass function of the random variable X denoting the sum of the numbers on the upper faces in random experiment of throwing two dice together.
8. Let X denotes the number of red balls when five balls are drawn at random with replacement from a bag containing 8 red balls and 6 green balls. Find the probability mass function of X .
9. If we roll a die and let X denote the number of rolls until the first 6 appears. Find the probability mass function of X .
10. Find the probability mass function of X denoting the number of girls in a family with three children.
11. From a lot of 10 items containing 3 defectives, a sample of 4 items is drawn at random. Let the random variable X denote the number of defective items in the sample. If the sample is drawn without replacement, then find-
 - (i) the pmf of X .
 - (ii) $P(X < 2)$
 - (iii) $P(1 \leq X < 3)$.
12. A random variable X has the following probability distribution-

X	0	1	2	3	4	5	6	7
$f(X = x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2 + k$

Find- (i) k (ii) $P(X < 6)$ (iii) $P(X \geq 5)$ (iv) $P(1 \leq X < 4)$ (v) If $P(X \leq a) > \frac{1}{2}$, find the minimum value of a .

13. A dealer in refrigerators estimates from his past experience, the probabilities of number of refrigerators X sold in a day during a week as follows-

X	0	2	5	8	9	10	12
p	0.03	0.20	0.23	0.12	0.22	0.05	0.15

Find the expected number of refrigerators sold in a day and the standard deviation.

14. The monthly demand for transistors is known to have the following probability distribution-

<i>Demand</i>	1	2	3	4	5	6
<i>Probability</i>	0.10	0.15	0.20	0.25	0.18	0.12

Find the expected demand for transistors and the standard deviation.

15. Find the distribution function of the random variable X that has the *pdf* as $f(x) = \frac{x}{15}, x = 1, 2, 3, 4, 5$.

16. If X has the distribution function

$$F(x) = \begin{cases} 0, & x < 1 \\ 1/3, & 1 \leq x < 4 \\ 1/2, & 4 \leq x < 6 \\ 5/6, & 6 \leq x < 10 \\ 1, & x \geq 10 \end{cases}$$

find- (i) $P(2 < X \leq 6)$ (ii) $P(X = 4)$ (iii) the probability distribution of X .

17. If X has the distribution function

$$F(x) = \begin{cases} 0, & x < -1 \\ 1/4, & -1 \leq x < 1 \\ 1/2, & 1 \leq x < 3 \\ 3/4, & 3 \leq x < 5 \\ 1, & x \geq 5 \end{cases}$$

find using both *pdf* and *cdf*— (i) $P(X \leq 3)$ (ii) $P(X = 3)$ (iii) $P(X \geq 3)$ (iv) $P(-1.4 < X < 4)$.

18. An investment firm offers its customers municipal bonds that mature after varying numbers of years. Given that the cumulative distribution function of T , the number of years to maturity for a randomly selected bond, is

$$F(t) = \begin{cases} 0, & t < 1 \\ 1/4, & 1 \leq t < 3 \\ 1/2, & 3 \leq t < 5 \\ 3/4, & 5 \leq t < 7 \\ 1, & t \geq 7 \end{cases}$$

Find-

- (i) $P(T = 5)$
(ii) $P(T > 3)$
(iii) $P(1.4 < t < 6)$.

19. Show that the function $f(x) = \frac{1}{\pi(1+x^2)}, -\infty < x < \infty$ is the probability density function

for a random variable X .

20. Show that the function $f(x) = \begin{cases} \frac{x^2}{2}, 0 \leq x < 1 \\ \frac{x^2 - 3(x-1)^2}{2}, 1 \leq x < 2 \\ \frac{x^2 - 3(x-1)^2 + 3(x-2)^2}{2}, 2 \leq x < 3 \\ 0, \text{elsewhere} \end{cases}$ is the probability density function for a random variable X .

21. The amount of bread (in hundreds of kilogram) X that a certain bakery is able to sell in a day is found to be a numerical valued random phenomenon with a probability function specified by

$$f(x) = \begin{cases} Ax, 0 \leq x < 5 \\ A(10-x), 5 \leq x < 10 \\ 0, \text{otherwise} \end{cases}$$

Find the value of A such that $f(x)$ is the probability density function. What is the probability that the amount of bread that will be sold tomorrow is (i) more than 500 kgs (ii) between 250 and 750 kgs (iii) less than 250 kgs.

22. Find the distribution function of the random variable X whose probability density is given by

$$\begin{aligned} \text{- (i) } f(x) &= \begin{cases} x, 0 < x < 1 \\ 2-x, 1 \leq x < 2 \\ 0, \text{elsewhere} \end{cases} \text{ and find } P(0.8 < X < 1.2). \\ \text{(ii) } f(x) &= \begin{cases} x/2, 0 < x \leq 1 \\ \frac{1}{2}, 1 \leq x < 2 \\ \frac{3-x}{2}, 2 < x < 3 \\ 0, \text{elsewhere} \end{cases} \text{ and find } P(-1/2 < X < 1/2), P(2 < X < 3). \end{aligned}$$

23. A continuous random variable has a *pdf* as $f(x) = 3x^2, 0 \leq x \leq 1$. Find a and b such that (i) $P(X \leq a) = P(X > a)$. (ii) $P(X > b) = 0.05$

24. A man travels to work by a car and the journey time T hours, has a *pdf* given by

$$f(t) = \begin{cases} 10ct^2, 0 \leq t < 0.6 \\ 9c(1-t), 0.6 \leq t < 1 \\ 0, \text{otherwise} \end{cases}$$

where c is a constant. Find the *cdf* and the probability that the journey time will be (i) more than 48 mins. (ii) between 24 and 48 mins.

25. The *cdf* of a continuous random variable is given by

$$F(x) = \begin{cases} 0, x < 0 \\ x^2, 0 \leq x < 1/2 \\ 1 - \frac{3}{25}(3-x)^2, 1/2 \leq x < 3 \\ 1, x \geq 3 \end{cases}$$

Find *pdf* of X and evaluate $P(|X| \leq 1)$ and $P(1/3 \leq X < 4)$ using both *pdf* and *cdf*.

26. The proportion of the budget for a certain type of industrial company that is allotted to environmental and pollution control is coming under scrutiny. A data collection project determines that the distribution of these proportions is given by-

$$f(x) = \begin{cases} 4(1+x), & 0 \leq x < 1/4 \\ \frac{2}{9}(1-4x^2), & 1/4 \leq x < 1 \\ 0, & \text{otherwise} \end{cases}$$

Find the distribution function and using it determine the probability that a company chosen at random expends more than 20% but less than 35% its budget on environmental and pollution controls?

27. Suppose that the time X in minutes that a person has to wait at a certain station for a train is found to be a random phenomenon with a probability function specified by the distribution function(*cdf*)-

$$F(x) = \begin{cases} 0, & x < 0 \\ \frac{x}{2}, & 0 \leq x < 1 \\ \frac{1}{2}, & 1 \leq x < 2 \\ \frac{x}{4}, & 2 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$$

Find the density function (*pdf*) and the probability that a person will have to wait between 1 and 3 minutes using both *pdf* and *cdf*.

28. The density function for a random variable X is given by-

$$f(x) = \begin{cases} x+1, & -1 \leq x < 0 \\ 1-x, & 0 \leq x < 1 \\ 0, & \text{elsewhere} \end{cases}$$

Find (i) the cumulative distribution function (ii) $P(-0.4 < X \leq 0.75)$ using both *pdf* and *cdf*.

29. The density function $f(t)$ for the age T years of a person is determined from long records and can be assumed to be-

$$f(t) = \begin{cases} 3 \times 10^{-9}t^2(100-t)^2, & 0 \leq t \leq 100 \\ 0, & \text{elsewhere} \end{cases}$$

Determine the probability that (i) the person will die between the ages 60 and 70 years. (ii) he will die between the ages 65 and 75, assuming he lived upto 60.

30. Suppose that the life length of a certain radio tube (in hours) is a continuous random variable with *pdf*

$$f(x) = \begin{cases} \frac{100}{x^2}, & x > 100 \\ 0, & \text{elsewhere} \end{cases}$$

(i) What is the probability that the tube will last less than 200h, if it is known that the tube is still functioning after 150h of service?

(ii) What is the probability that if 3 of such tubes are installed in a set, exactly one will have to be replaced after 150h of service?

(iii) What is the maximum number of tubes may be inserted into a set so that there is a probability of 0.1 that after 150h of service all of them are still functioning?

31. Let X be a continuous random variable with *pdf* given by-

$$f(x) = \begin{cases} kx, & 0 \leq x < 1 \\ k, & 1 \leq x < 2 \\ -kx + 3k, & 2 \leq x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

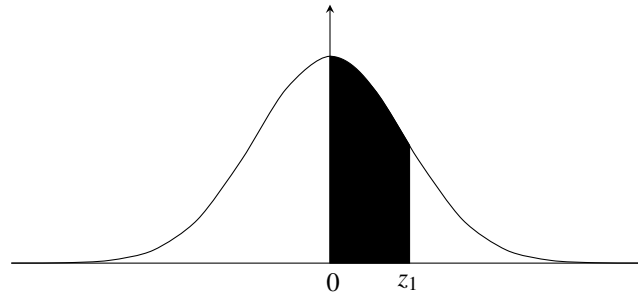
Determine

- (i) the constant k
- (ii) the *cdf*
- (iii) $P(1.75 < X < 2.25)$.

32. Consider the function-

$$f(x) = \begin{cases} \frac{x^2}{2}, & 0 \leq x < 1, \\ \frac{x^2 - k(x-1)^2}{2}, & 1 \leq x < 2, \\ \frac{x^2 - k(x-1)^2 + k(x-2)^2}{2}, & 2 \leq x < 3 \\ 0, & \text{elsewhere} \end{cases}$$

- (i) Find the value of k for which $f(x)$ is the density function of X .
- (ii) Find the distribution function of X .
- (iii) Find $P(0.25 < X < 2.25)$ using both *pdf* and *cdf*.

Figure 3.2: $P(0 \leq Z \leq z_1) = \text{Area under the standard normal curve for } 0 \leq z \leq z_1$

Area under the standard normal curve for $0 \leq z \leq z_1$										
z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.0000	0.0040	0.0080	0.0120	0.0160	0.0199	0.0239	0.0279	0.0319	0.0359
0.1	0.0398	0.0438	0.0478	0.0517	0.0557	0.0596	0.0636	0.0675	0.0714	0.0753
0.2	0.0793	0.0832	0.0871	0.0910	0.0948	0.0987	0.1026	0.1064	0.1103	0.1141
0.3	0.1179	0.1217	0.1255	0.1293	0.1331	0.1368	0.1406	0.1443	0.1480	0.1517
0.4	0.1554	0.1591	0.1627	0.1664	0.1700	0.1736	0.1772	0.1808	0.1844	0.1879
0.5	0.1915	0.1950	0.1985	0.2019	0.2054	0.2088	0.2123	0.2157	0.2190	0.2224
0.6	0.2257	0.2291	0.2324	0.2357	0.2389	0.2422	0.2454	0.2486	0.2518	0.2549
0.7	0.2580	0.2612	0.2642	0.2673	0.2704	0.2734	0.2764	0.2794	0.2823	0.2852
0.8	0.2882	0.2910	0.2939	0.2967	0.2996	0.3023	0.3051	0.3079	0.3106	0.3133
0.9	0.3159	0.3186	0.3212	0.3238	0.3264	0.3290	0.3315	0.3340	0.3365	0.3389
1.0	0.3414	0.3438	0.3461	0.3485	0.3508	0.3531	0.3554	0.3577	0.3599	0.3621
1.1	0.3643	0.3665	0.3686	0.3708	0.3729	0.3749	0.3770	0.3790	0.3810	0.3830
1.2	0.3849	0.3869	0.3888	0.3906	0.3925	0.3943	0.3962	0.3980	0.3997	0.4015
1.3	0.4032	0.4049	0.4066	0.4082	0.4099	0.4115	0.4131	0.4146	0.4162	0.4177
1.4	0.4192	0.4207	0.4222	0.4236	0.4251	0.4265	0.4278	0.4292	0.4306	0.4319
1.5	0.4332	0.4345	0.4357	0.4370	0.4382	0.4394	0.4406	0.4418	0.4429	0.4441
1.6	0.4452	0.4453	0.4474	0.4484	0.4495	0.4505	0.4515	0.4525	0.4535	0.4545
1.7	0.4554	0.4564	0.4573	0.4582	0.4591	0.4599	0.4608	0.4610	0.4625	0.4633
1.8	0.4641	0.4648	0.4656	0.4664	0.4671	0.4678	0.4686	0.4693	0.4699	0.4706
1.9	0.4713	0.4719	0.4726	0.4732	0.4738	0.4744	0.4750	0.4756	0.4762	0.4767
2.0	0.4773	0.4778	0.4783	0.4788	0.4793	0.4798	0.4803	0.4808	0.4812	0.4817
2.2	0.4821	0.4826	0.4830	0.4834	0.4838	0.4842	0.4846	0.4850	0.4854	0.4857
2.3	0.4893	0.4896	0.4898	0.4901	0.4904	0.4906	0.4909	0.4911	0.4914	0.4916
2.4	0.4918	0.4920	0.4922	0.4925	0.4927	0.4929	0.4931	0.4933	0.4934	0.4936
2.5	0.4938	0.4940	0.4941	0.4943	0.4945	0.4946	0.4948	0.4949	0.4951	0.4952
2.6	0.4953	0.4955	0.4956	0.4957	0.4959	0.4960	0.4961	0.4962	0.4963	0.4964
2.7	0.4965	0.4966	0.4967	0.4968	0.4969	0.4970	0.4971	0.4972	0.4973	0.4974
2.8	0.4974	0.4975	0.4976	0.4977	0.4977	0.4978	0.4979	0.4979	0.4980	0.4981
2.9	0.4981	0.4982	0.4982	0.4983	0.4984	0.4984	0.4985	0.4985	0.4986	0.4986
3.0	0.4986	0.4987	0.4987	0.4988	0.4988	0.4988	0.4989	0.4989	0.4990	0.4990

Exercise 3.2

1. Let X be a normal variate with mean 12 and SD 4, find-
(i) $P(X \geq 20)$ (ii) $P(10 \leq X \leq 15)$ (iii) $P(X \leq 18)$ (iv) $P(|X - 8| > 5)$.
2. Let Z be a standard normal variate. Determine the value of t if-
(i) $P(0 \leq Z \leq t) = 0.4236$ (ii) $P(Z \leq t) = 0.7967$ (iii) $P(t \leq Z \leq 2) = 0.1000$.
3. If X is a normal variate with mean 11 and standard deviation 1.5, find the number a such that $P(X > a) = 0.09$.
4. If skulls are classified as A, B, C according as the length-breadth index is under 75, between 75 and 80, over 80, find approximately (assuming that the distribution is normal) the mean and standard deviation of a series in which A are 58%, and C are 4%. Also find the % of skulls classified as B.
5. A manufacturer does not know the mean and the standard deviation of the diameters of the ball bearings he is producing. However the system rejects all the bearings with diameters larger than 2.4cms and those with diameters under 1.8cms. Out of 1000 ball bearings, 8% are rejected as too small and 5.5% as too big.
(i) What are the mean and the standard deviations of the diameters of the ball bearings.
(ii) Find the number of ball bearings with diameters between 1.95cm and 2.25cms.
6. Most graduate schools of business require applicants for admission to take the Graduate Management Admission Council's GMAT examination. Scores on the GMAT are roughly normally distributed with a mean of 547 and a standard deviation of 102.
(i) What is the probability of an individual scoring above 450 on the GMAT?
(ii) What is the highest score of an individual scoring below top 10% on the GMAT?
(iii) What is the least score of an individual on the GMAT scoring in the highest 2%?
7. In a normal distribution, 7% of the items are under 35 and 89% are under 63. Find the mean and the SD of the distribution.
8. It is known that the life length (years) of a certain type of storage battery follows a normal distribution. If 22% of these lasts for less than 2.5 years and 15% of them lasts for more than 5 years, find the % of the batteries lasting between 1.5 and 2.25 years.
9. In an engineering examination, a student is considered to have failed, secured second class, first class and distinction, according as the score is less than 45%, between 45% and 60%, between 60% and 75% and above 75% respectively. In a particular year, 10% of the students failed in the examination and 5% got distinction. Assuming the normal distribution of the marks, find the percentages of the students who got first class and second class.
10. Entry to a certain University is determined by a national test. The scores on this test are normally distributed with a mean of 530 and a standard deviation of 100. A student wants to be admitted to this university and she knows that she must score better than at least 70% of the students who took the test. She takes the test and scores 585. Will she be admitted to this university?
11. The life lengths (in hours) of two electronic devices A and B have normal distributions $N(44, 36)$ and $N(45, 9)$ respectively. If the electronic device is to be used for a 47 – hours period, which device is to be preferred?

12. A random variable has a normal distribution with $\sigma = 10$. If the probability that the random variable will take on a value less than 82.5 is 0.8212, what is the probability that it will take a value greater than 58.3?
13. The weight of a sophisticated running shoe is normally distributed with a mean of 12 ounces and a standard deviation of 0.5 ounce. (i) What is the probability that a shoe weighs more than 12.4 ounces? (ii) What must the standard deviation of weight be in order for the company to state that 97% of its shoes weighs less than 13 ounces? (iii) What must the mean weight be for the company to state that only 5% of its shoes weighs more than 12.5 ounces?
14. It is known from the past experience that the number of telephone calls made daily in a certain community between 3 P.M. and 4 P.M. is normally distributed with mean 352 and a standard deviation of 31. Find the percentage of time for which there are (i) more than 400 calls (ii) between 250 and 375 calls.
15. The local authorities in a certain city install 10000 electric lamps in the streets of the city. If these lamps have an average life of 1000 burning hours with a standard deviation of 200 burning hours, how many lamps might be expected to fail in (i) first 800 burning hours (ii) between 250 and 425 burning hours? After how many burning hours would you expect 10% of the lamps to fail?
16. Replacement times for *CD* players are normally distributed with a mean of 7.1 years and a standard deviation of 1.4 years (data from consumer reports). If you are the manufacturer and want to provide a warranty such that 98% of the players need replacement after the warranty expires, how long should the warranty period be?
17. The speed of a file transfer from a server on campus to a personal computer at a student's home on a weekday evening is normally distributed with a mean of μ kilobits per second and a standard deviation of σ kilobits per second. If the probability that the file will transfer at a speed of 70 kilobits per second or more is 0.85 and the probability that the file will transfer at a speed of less than 85 kilobits per second is 0.21, find μ and σ .
18. The lifetimes of certain electronic devices have a mean of 300 hours and standard deviation of 25 hours. Assuming that the distribution of these lifetimes, which are measured to nearest hour, can be approximated closely with a normal curve, find- (i) the probability that any one of these electronics devices will have a lifetime of more than 350 hours. (ii) what percentage will have lifetime of 300 hours and less? (iii) what percentage will have lifetime from 2203 hours to 260 hours?
19. The mean inside diameter of a sample of 500 washers produced by a machine is 5.02 *mm* and standard deviation is 0.05 *mm*. The purpose for which these washers are intended allows a maximum tolerance in the diameter of 4.96 to 5.08 *mm*, otherwise the washers are considered defective. Determine the percentage of defective washers produced by the machine, assuming the diameters are normally distributed.
20. If the diameters of ball bearings are normally distributed with mean 0.614 inches and standard deviation 0.0025 inches, determine the percentage of ball bearings with diameters (i) between 0.610 and 0.618 inches (ii) greater than 0.617 inches (iii) less than 0.608 inches (iv) equal to 0.615 inches.
21. The diameters of shafts produced in a factory confirms to normal distribution. 31% of the shafts have a diameter less than 45 *mm* and 8% have more than 64 *mm*. Find the mean and the standard deviation of the diameter of the shafts.

Exercise 4.1

1. Consider the random process $X(t) = A \cos(wt + \theta)$, where A and θ are independent and uniformly distributed random variables in the intervals $(-k, k)$ and $(-\pi, \pi)$ respectively. Find the mean, auto correlation, auto co-variance, variance and correlation of $X(t)$.
2. Let $X(t)$ be a random process with mean 3 and the auto correlation given by

$$R(t_1, t_2) = 9 + 4e^{-0.2|t_1 - t_2|}$$

Determine the mean, auto correlation, auto co-variance, variance and correlation of the random variables $Z = X(5)$ and $W = X(8)$.

3. Consider the random process $X(t) = a \cos \lambda t + b \sin \lambda t$, where a and b are independent random variables with $E(a) = E(b) = 0$, $Var(a) = Var(b) = \sigma^2$ and θ is a constant. Obtain the mean, variance and the auto-correlation of the process.
4. A random process is defined by $X(t) = Y + 1 - t$, where Y is a uniformly distributed random variable in $(0, 1)$. Find (i) cdf of $X(t)$ (ii) $E[X(t)]$ and $C(t_1, t_2)$.
5. In the fair-coin experiment, we define the process $X(t)$ as follows:

$$X(t) = \begin{cases} \sin \pi t, & \text{if head shows} \\ 2t, & \text{if tail shows} \end{cases}$$

Find (i) $E\{X(t)\}$ and (ii) $F(x, t)$ for $t = 0.25, 0.5, 1$.

6. In the fair-coin experiment, we define the process $X(t)$ as follows:

$$X(t) = \begin{cases} 1 - t^2, & \text{if head shows} \\ 4t + 1, & \text{if tail shows} \end{cases}$$

Find (i) $E\{X(t)\}$ and (ii) $F(x, t)$ for $t = 0.5, 0.75$.

Answers

1. $\mu_X(t) = 0, R(t_1, t_2) = \frac{k^2}{6} \cos(w(t_1 - t_2)), C(t_1, t_2) = \frac{k^2}{6} \cos(w(t_1 - t_2)),$
 $\sigma_X^2(t) = \frac{k^2}{6}, \rho(t_1, t_2) = \cos(w(t_1 - t_2))$
2. $E[Z] = E[X(5)] = 3, E[W] = E[X(8)] = 3, Cov(Z, W) = 4e^{-0.6}, Var(Z) = Var(W) = 4,$
3. $\mu_X(t) = 0, Var(X(t)) = \sigma^2, R(t_1, t_2) = \sigma^2 \cos \lambda(t_1 - t_2)$
4. (i) $F_X(t) = \begin{cases} 0, & x \leq 1 - t \\ x + t - 1, & 1 - t < x < 2 - t \\ 1, & x \geq 2 - t \end{cases}$ (ii) $E[X(t)] = \frac{3}{2} - t, C(t_1, t_2) = \frac{1}{12}$
5. $X(0.5) = \begin{cases} 1, & \text{If head shows} \\ 1, & \text{If tail shows} \end{cases}$
6. $E\{X(0.5)\} = 3.75, E\{X(0.75)\} = 2.23,$
 $F(x, 0.5) = \begin{cases} 0, & x < 0.75 \\ \frac{1}{2}, & 0.75 \leq x < 3 \\ 1, & x \geq 3 \end{cases}, F(x, 0.75) = \begin{cases} 0, & x < 0.44 \\ \frac{1}{2}, & 0.44 \leq x < 4 \\ 1, & x \geq 4 \end{cases}$

Exercise 5.2

1. A salesman's territory consists of cities A, B and C . He never sells in the same city on successive days. If he sells in city A , then next day he sells in city B . However if he sells on either B or C , then next day he is twice as likely to sell in city A as in the other city. Find the transition probability matrix.
2. An auto insurance company classifies its customers in three categories: poor, satisfactory and preferred. No one moves from poor to preferred or from preferred to poor in one year. 40% of the customers in the poor category become satisfactory, 30% of those in the satisfactory category moves to preferred, while 10% become poor; 20% of those in the preferred category are downgraded to satisfactory. If currently, there are 22%, 48% and 30% customers in the three categories respectively, find the % of the customers in each category (i) after three years (ii) in the long run.
3. A manufacturing company has a certain piece of equipment that is inspected at the end of each day and classified as just overhauled, good, fair or inoperative. If the piece is inoperative, it is overhauled, a procedure that takes one day. Assume that the working condition of the equipment follows a Markov process with the following transition matrix-

$$P = \begin{bmatrix} 0 & 3/4 & 1/4 & 0 \\ 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1/2 & 1/2 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

It costs 125/– to overhaul a machine (including lost time) on an average and 75/– in production lost if the machine is found to be inoperative. Compute the expected per day cost of maintenance (i) after 4 days if the currently the state distribution of the equipment is $[0.25 \ 0.22 \ 0.26 \ 0.27]$ (ii) in the long run.

4. The school of international studies for population found out by its survey that the mobility of the population of a state to village, town and a city is in the following percentages-

		To		
		Village	Town	City
From	Village	50	30	20
	Town	10	70	20
	City	10	40	50

What will be the proportion of population in village town and the city after two years, given that the present population has proportions of 0.7, 0.2 and 0.1 in the village, town and city respectively? What will be the respective proportions in the long run?

5. A salesman's territory consists of cities A, B and C . He never sells in the same city on successive days. If he sells in city A , then next day he sells in city B . However if he sells on either B or C , then next day he is twice as likely to sell in city A as in the other city. In the long run how often does he sell in each of the cities?
6. A professor has three pet questions, one of which occurs on every test he gives. The students know his habit well. He never uses the same question twice in a row. If he used question 1 last time, he tosses a coin and uses the question 2 if a head comes up. If he used question 2 he tosses two coins and switches to question 3 if both come up with head. If he used question 3, he tosses three coins and switches to question 1 if all three come

up with head. In the long run which question does he use most often and with how much frequency is it used?

7. A grocer stocks his store with three types of detergents A, B and C. When brand A is sold out the probability is 0.7 that he stocks up with brand A again. When he sells out brand B the probability is 0.8 that he will stock up again with brand B. Finally when he sells out brand C, the probability is 0.6 that he will stock up with brand C again. When he switches to another detergent he does so for the remaining two brands. Find the transition matrix. In the long run, how does he with equal probabilities stock up with detergent?
8. A communication source can generate one of three possible messages 1, 2 and 3 with the following transition probabilities-

		Next Message		
		1	2	3
Current Message	1	0.5	0.3	0.2
	2	0.4	0.2	0.4
	3	0.3	0.3	0.4

Initially, the probabilities of generating the messages 1, 2 and 3 are 0.3, 0.3 and 0.4 respectively. What are the probabilities after 3 periods.

9. A man is at an integral point on the x -axis between the origin and the point 3. He takes a unit step to the right with probability $1/3$ or to the left with probability $2/3$, unless he is at the origin, where he takes a step to the right to reach the point 1 or is at the point 3, where he takes a step to the left to reach the point 2. What is the probability that he is at the point 1 after 3 walks?.
10. A gambler's luck follows a pattern. If he wins a game, the probability of his winning the next game is 0.6. However if he loses a game, the probability of his losing the next game is 0.7. There is an even chance that the gambler wins the first game. What is the probability that the gambler wins (i) the second game (ii) the third game?
11. Assume that the weather in a certain locality can be modeled as a Markov chain whose transition probabilities are given below-

Today's weather	Tomorrow's weather		
	Fair	Cloudy	Rainy
Fair	0.80	0.15	0.05
Cloudy	0.50	0.30	0.20
Rainy	0.60	0.30	0.10

If the initial state distribution is given by $p^{(0)} = [0.70 \ 0.20 \ 0.10]$, find $p^{(3)}$ and the steady state distribution.

12. A professor tried not to be late for class too often. If he is late one day, he is 90% sure to be on time next day. If he is on time, then the next day there is a 30% chance of his being late. In the long run how often he is late for the class?
13. On January 1 this year bakery A had 40% of its local market share while the other two bakeries B and C had 40% and 20 % of the market share respectively. Based upon a study by a marketing research firm, the following facts were compiled- Bakery A retains 90% of its customers while gaining 5% of B's customers and 10% of C's customers. Bakery B retains 85% of its customers while gaining 5% of A's customers and 7% of C's customers.

Bakery C retains 83% of its customers and gains 5% of A's customers and 10% of B's customers. What will each firm's share be on January 1 next year and what will each firm's market share be at equilibrium?

14. In a certain region, the weather patterns have the following sequence. A day is described as sunny (S) if the sun shines for more than 50% of daylight hours and cloudy (C) if the sun shines for less than 50% of daylight hours. Data indicate that if it is cloudy one day then it is equally likely to be cloudy or sunny on the next day; if it is sunny there is a probability $1/3$ that it is cloudy and $2/3$ that it is sunny the next day.
- (i) If it is cloudy today, what are the probabilities that it is (a) cloudy, (b) sunny, in three days' time?
- (ii) Compute P^5 and P^{10} . How do you think that P^n behaves as $n \rightarrow \infty$?
- (iii) How does the state probability vector $p^{(n)}$ behave as $n \rightarrow \infty$?
15. Show that If P is the tpm of a homogeneous Markov chain, then the n step tpm is given by

$$P^{(n)} = P^n = P^{n-1} \times P.$$

Answers

1. $P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix} \end{matrix}$
2. (i) 3.2%, 39.4% and 47.3% (ii) $x = \frac{1}{11} = 0.09, y = \frac{4}{11} = 0.36, z = \frac{6}{11} = 0.55$
3. (i) 37.5% – (ii) 36.4% –
4. 25.2%, 47.9%, 26.9% and $x = 0.17, y = 0.55$ and $z = 0.28$.
5. In city A $\frac{8}{20}$, in B $\frac{9}{20}$, in C $\frac{3}{20}$.
6. Question no 2 with the frequency 40%.
7. $[0.31 \ 0.44 \ 0.25]$, In the long run he stocks the three detergents in proportion of 31%, 46%, 23%
8. 0.41, 0.27 and 0.32
9. $\frac{11}{27}$
10. (i) 0.48, 0.52 and (ii) 0.4, 0.56
11. $[0.73 \ 0.19 \ 0.08]$ and $[0.73 \ 0.19 \ 0.08]$.
12. In the long run he is late for the class 25% of the time.
13. 40%, 37.4%, 22.6% and 43%, 28%, 29%
14. (i) $\frac{29}{42}$ and $\frac{43}{72}$ (ii) $P^{(n)} \rightarrow \begin{bmatrix} 0.40 & 0.60 \\ 0.40 & 0.60 \end{bmatrix}$ as $n \rightarrow \infty$. (iii) $[0.4 \ 0.6]$

Exercise 5.1

1. A salesman's territory consists of cities A, B and C . He never sells in the same city on successive days. If he sells in city A , then next day he sells in city B . However if he sells on either B or C , then next day he is twice as likely to sell in city A as in the other city. Find the transition probability matrix.
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It costs 125/– to overhaul a machine (including lost time) on an average and 75/– in production lost if the machine is found to be inoperative. Compute the expected per day cost of maintenance (i) after 4 days if the currently the state distribution of the equipment is $[0.25 \ 0.22 \ 0.26 \ 0.27]$ (ii) in the long run.

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What will be the proportion of population in village town and the city after two years, given that the present population has proportions of 0.7, 0.2 and 0.1 in the village, town and city respectively? What will be the respective proportions in the long run?

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		Next Message		
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Current Message	1	0.5	0.3	0.2
	2	0.4	0.2	0.4
	3	0.3	0.3	0.4

Initially, the probabilities of generating the messages 1, 2 and 3 are 0.3, 0.3 and 0.4 respectively. What are the probabilities after 3 periods.

9. A man is at an integral point on the x -axis between the origin and the point 3. He takes a unit step to the right with probability $1/3$ or to the left with probability $2/3$, unless he is at the origin, where he takes a step to the right to reach the point 1 or is at the point 3, where he takes a step to the left to reach the point 2. What is the probability that he is at the point 1 after 3 walks?.
10. A gambler's luck follows a pattern. If he wins a game, the probability of his winning the next game is 0.6. However if he loses a game, the probability of his losing the next game is 0.7. There is an even chance that the gambler wins the first game. What is the probability that the gambler wins (i) the second game (ii) the third game?
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Today's weather	Tomorrow's weather		
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Bakery C retains 83% of its customers and gains 5% of A's customers and 10% of B's customers. What will each firm's share be on January 1 next year and what will each firm's market share be at equilibrium?

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- (i) If it is cloudy today, what are the probabilities that it is (a) cloudy, (b) sunny, in three days' time?
- (ii) Compute P^5 and P^{10} . How do you think that P^n behaves as $n \rightarrow \infty$?
- (iii) How does the state probability vector $p^{(n)}$ behave as $n \rightarrow \infty$?
15. Show that If P is the tpm of a homogeneous Markov chain, then the n step tpm is given by

$$P^{(n)} = P^n = P^{n-1} \times P.$$

Answers

1. $P = \begin{matrix} & \begin{matrix} A & B & C \end{matrix} \\ \begin{matrix} A \\ B \\ C \end{matrix} & \begin{bmatrix} 0 & 1 & 0 \\ 2/3 & 0 & 1/3 \\ 2/3 & 1/3 & 0 \end{bmatrix} \end{matrix}$
2. (i) 3.2%, 39.4% and 47.3% (ii) $x = \frac{1}{11} = 0.09, y = \frac{4}{11} = 0.36, z = \frac{6}{11} = 0.55$
3. (i) 37.5% – (ii) 36.4% –
4. 25.2%, 47.9%, 26.9% and $x = 0.17, y = 0.55$ and $z = 0.28$.
5. In city A $\frac{8}{20}$, in B $\frac{9}{20}$, in C $\frac{3}{20}$.
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11. $[0.73 \ 0.19 \ 0.08]$ and $[0.73 \ 0.19 \ 0.08]$.
12. In the long run he is late for the class 25% of the time.
13. 40%, 37.4%, 22.6% and 43%, 28%, 29%
14. (i) $\frac{29}{42}$ and $\frac{43}{72}$ (ii) $P^{(n)} \rightarrow \begin{bmatrix} 0.40 & 0.60 \\ 0.40 & 0.60 \end{bmatrix}$ as $n \rightarrow \infty$. (iii) $[0.4 \ 0.6]$

Exercise 5.3

1. On the average a submarine on patrol sights 6 enemy ships per hour. Assuming that the number of ships sighted in a given length of time is a Poisson variate, find the probability of sighting- (i) at least 3 ships in the next 20 min.
(ii) atleast 3 but not more than 6 ships in next one and half hour.
(iii) at the most 4 ships in 15 mins.
2. Suppose that the customers arrive at a bank counter according to a Poisson process with a mean rate of 3 per minute. Find the probability that during a time interval of 2 minutes (i) exactly 4 customers arrive and (ii) more than 4 customers arrive.
3. If the customers arrive at a counter in accordance with a Poisson process with a mean rate of 2 per minute, find the probability that the interval between the two consecutive arrivals is (i) more than 1 min. (ii) between 1 and 2 min. (iii) 4 min or less.
4. A machine goes out of order, whenever a component fails. The failure of this part follows a Poisson process with a mean rate of 1 per week. Find the probability that 2 weeks have elapsed since last failure. If there are 5 spare parts of this component in an inventory and that the next supply is not due in 10 weeks, find the probability that the machine will not fail in the next 10 weeks.
5. Patients arrive randomly and independently at a doctor's consulting room from 8 : 00 am at an average rate of one in 5 min. The waiting room can hold 12 persons. What is the probability that the room will be full when the doctor arrives at 9 : 00 am?
6. On the average a submarine on patrol sights 6 enemy ships per hour. Assuming that the number of ships sighted in a given length of time is a Poisson variate, find the probability of sighting-(i) 6 ships in the next half-an-hour (ii) 4 ships in the next 2 hour (iii) at least 1 ship in the next 15 min and (iv) at least 2 ships in the next 20 min.
7. Messages arrive at a telegraph office in accordance with the laws of a Poisson process with a mean rate of 3 messages per hour. What is the (i) probability that no message will have arrived during the morning hours i.e. between 8 : 00 am and 12 : 00 noon? (ii) the distribution of the time at which the first afternoon message arrives?
8. Assume that a circuit has an IC whose time to failure is a exponentially distributed RV with expected lifetime of 3 months. If there are 10 spare IC 's and time from failure to replacement is zero, what is the probability that (i) 5 IC 's failed in first 4 months (ii) the circuit is operational only for 8 months (iii) the circuit can be kept operational for at least 1 year?
9. A radioactive source emits particles at a rate of 5 per minute in accordance with Poisson process. Each particle emitted has a probability of 0.6 of being recorded. Find the probability that 10 particles are recorded in 4 – min period.
10. A radioactive source emits particles at a rate of 6 per minute in accordance with Poisson process. Each particle emitted has a probability $1/3$ of being recorded. Find the probability that (i) at least 5 particles are recorded. (ii) exactly 3 particle are recorded (iii) at the most 4 particles are recorded in a 5–minute period.
11. Suppose that customers arrive at a counter independently from 2 different sources. Arrivals occur in accordance with a Poisson process with mean rate of 6 per hour from the first source and 4 per hour from the second source. Find the mean interval between any two consecutive arrivals.

12. Passengers arrive at a terminal for boarding the next bus. Their arrivals are Poisson with an average arrival rate of 1 per minute. The times of departure of each bus are Poisson with an average departure rate of 2 per hour. Assume that the capacity of the bus is large. Find the average number of passengers in (i) each bus (ii) the first bus that leaves after 9 A.M.
13. Passengers arrive at a terminal after 9 A.M. The times of their arrivals are Poisson with an mean density $\lambda = 1$ per minute. The time interval from 9 A.M. to the departure of the next bus is $RV\ T$. Find the mean number of passengers in this bus (i) if T has an exponential density with mean 30 min and (ii) if T is uniform between 0 and 60 min.

Answers

- | | |
|--|--------------------------------|
| 1. (i)0.32 (ii)0.20 (iii)0.98 | 8. (i)0.01 (ii)0.00 (iii)1 |
| 2. (i)0.13 (ii)0.72 | 9. 0.10 |
| 3. (i)0.14 (ii)0.23 (iii)1 | 10. (i)0.97 (ii)0.01 (iii)0.03 |
| 4. (i)0.14 and 0.07 | 11. 6 mins. |
| 5. 0.11 | 12. (i)30 (ii)30 |
| 6. (i)0.05 (ii)0.01 (iii)0.78 (iv)0.59 | 13. (i)30 (ii)30 |
| 7. (i) e^{-12} (ii) $1 - e^{-3(t-12)}$ | |

Exercise 6.1 Reliability

1. The density function of the time to failure in years of the gizmos manufactured by a certain company is given by

$$f(t) = \frac{200}{(t+10)^3}, t \geq 0.$$

(a) Derive the reliability function and determine the reliability for the first year of operation.

(b) Compute MTTF.

(c) What is the design life for a reliability of 0.95?

(d) Will a one year burn-in period improve the reliability in part (a)? If so, what is the new reliability?

2. The reliability of a turbine blade is given by $R(t) = (1 - \frac{t}{t_0})^2, 0 \leq t \leq t_0$, where t_0 is the maximum life of the blade.

(a) Show that the blades are experiencing wear out.

(b) Compute MTTF as a function of maximum life.

(c) If the maximum life is 2000 operating hours, what is the design life for a reliability of 0.90?

3. Given that-

$$R(t) = e^{-\sqrt{0.001t}}, t \geq 0$$

(i) Compute the reliability for 50 hours mission.

(ii) Show that the hazard rate is decreasing.

(iii) compute MTTF.

(iv) Given a 10 hour wear in period, compute the reliability for a 50 hour mission.

(v) What is the design life for a reliability of 0.95, given a 10 hour wear in period?

4. A one year guarantee is given based on the assumption that no more than 10% of the items will be returned. Assuming an exponential distribution, what is the maximum failure rate that can be tolerated?

5. A home computer manufacturer determines that his machine has a constant failure rate of $\lambda = 0.4$ per year in normal use.

(i) For how long should the warranty be set, if no more than 5% of the computers are to be returned to the manufacturer for repair?

(ii) Find the expected time to failure.

(iii) The design life corresponding to the reliability 0.82.

(iv) What is the reliability for next two years of a machine which is already used for 6 months?

6. A household appliance is advertised as having more than a 10 year life. If its *pdf* is given by $f(t) = 0.1(1 + 0.05t)^{-3}; t \geq 0$. Determine its reliability for the next 10 years, if it has survived a 1 year warranty period. What is the MTTF before the warranty period is over? What is its MTTF after the warranty period assuming that it has still survived?

7. The time to failure in operating hours of a critical solid state power unit has hazard function $\lambda(t) = 0.003(\frac{t}{500})^{0.5}, t \geq 0$.

(a) What is the reliability if the power unit must operate continuously for 50 hours?

(b) Determine the design life if a reliability of 0.90 is desired.

(c) Compute MTTF.

- (d) Given that the unit has operated for 50 hours, what is the probability that it will survive a second 50 hours of operation?
8. A density function of the time to failure of a new fuel injection system which experiences high failure rate is given by $f(t) = 1.5/(1+t)^{2.5}$ where t is measured in years. The reliability over its intended life of 2 years is unacceptable. Will a wear-in period of 6 months significantly improve upon this reliability? If so, by how much?
9. The failure distribution is defined by $f(t) = \frac{2}{(1+0.05t)^3}, t \geq 0$ hours.
- (i) What is the probability of failure within a 100-hour warranty period?
 - (ii) Compute $MTTF$.
 - (iii) Find the design life for a reliability of 0.90.
 - (iv) Given a 2 year burn-in period, compute the reliability for one year of operation.

Answers

1. (a) $R(t) = \frac{100}{(t+10)^2}, R(1) = \frac{100}{121} = 0.8264$ (b) 10 years (c) 0.2596 years or 95 days (d) $R(1/1) = \frac{121}{144} = 0.8403 > 0.8264$.
2. (b) $\frac{t_0}{3}$ (c) 102.63 hours.
3. (i) $R(50) = 0.80$ (iii) $MTTF = 200h$ (iv) $R(50/10) = 0.87$. (v) $t_D = 12.89$ hours.
4. 0.1054/year
5. (i) 47 days (ii) 2.5 years (iii) 181 days (iv) 0.45
6. 0.46, 19.05 years
7. (a) 0.97, (b) 111.50 hours (c) 451.38 hours (d) 0.94
8. Yes by 0.16
9. (i) 0.45 (ii) 400 hours (iii) 74.28 hours (iv) 0.44

Exercise 6.2 *Reliability of systems*

- The time to failure (in years) of a certain brand of computer has the density function $f(t) = 1/(1+t)^2, t > 0$. If three of these computers are placed parallel aboard the proposed space station then, what is the system reliability for-
 - the first 6 months of operation?
 - next 2 years after the system has already operated for 6 months?
- Thermo-couples of a particular design have a failure rate of 0.008 per hour. How many thermo-couples must be placed in parallel if the system is to run for 100 hours with a system failure probability of no more than 0.05? Assume that all failures are independent.
- If two identical components having a guaranteed life of 2 months and a constant failure rate of 0.15 per year are connected in parallel, what is the system reliability for 10,000 hours of continuous operation?
- Find the minimum number of redundant components each having a reliability of 0.4 necessary to achieve a system reliability of 0.95. There is a common mode failure probability of 0.03.
- It is known that the reliability function for a critical solid state power unit for use in a communication satellite is $R(t) = \frac{10}{10+t}, t \geq 0$ and t measured in years.
 - How many units must be placed in parallel, in order to achieve a reliability of 0.98 for 5 years operation?
 - If there is an additional common mode with constant failure rate of 0.002 as a result of environmental factors, how many units must be placed in parallel?
- The time to failure (in years) of a certain brand of computer has the density function $f(t) = 1/(1+t)^2, t \geq 0$ in years.
 - How many computers must be placed in parallel, in order to achieve a reliability of 0.85 for 5 years operation?
 - If there is an additional common mode with constant failure rate of 0.02 as a result of environmental factors, how many computers must be placed in parallel?
- Find the reliability of the system shown in the figure-

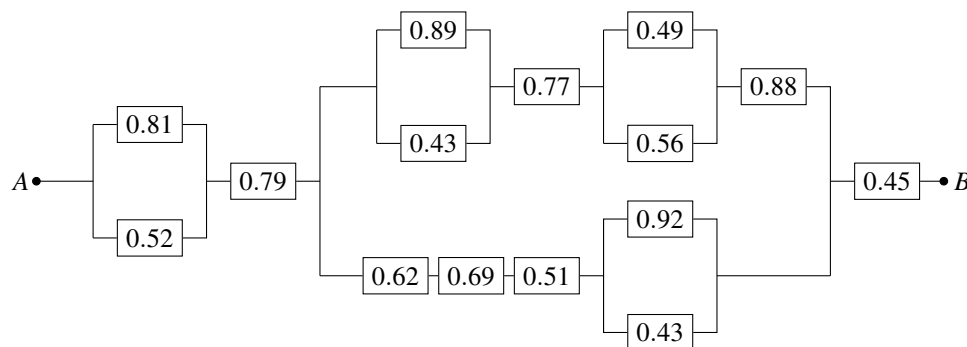


Figure 6.7

- Find the reliability of the system shown in the figure-

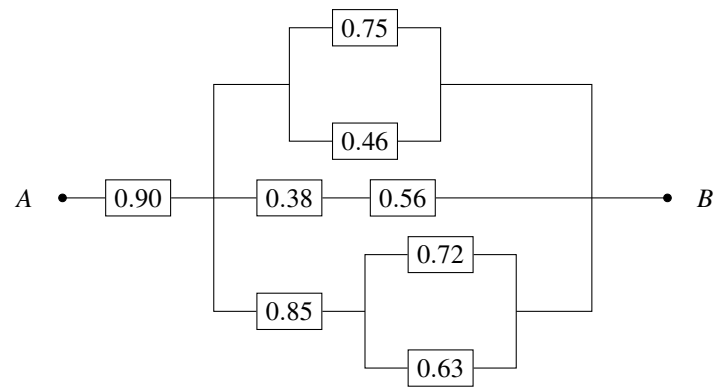


Figure 6.8

Answers

1. (i) 0.96 (ii) 0.66
2. 6 or more
3. 0.97
4. Atleast 8
5. (i) 4 or more (ii) 5 or more
6. (i) 10 or more (ii) 16 or more
7. 0.19
8. 0.88

Exercise 7.1 *Infinite capacity single server queuing model*

1. A branch office of a large engineering firm has one on-line terminal connected to a central computer system for 16 hours each day. Engineers, who work throughout the city, drive to the branch office to use the terminal for making routine calculations. The arrival pattern of engineers is random (Poisson) with an average of 20 persons per day using the terminal. The distribution of time spent by an engineer at the terminal is exponential with an average time of 30 minutes. The branch manager receives complaints from the staff about the length of time many of them have to wait to use the terminal. It does not seem reasonable to the manager to procure another terminal when the present one remains idle for some time. Find-
 - (i) Average number of engineers waiting in the queue.
 - (ii) Average number of engineers at the branch office.
 - (iii) Average waiting time in queue.
 - (iv) Average time an engineer spends at the branch office.
 - (v) Probability that the waiting time of an engineer exceeds 72 mins at the branch office.
2. In a departmental store one cashier is there to serve the customers. The customers pick up their needs by themselves. The arrival rate is 9 customers for every 5 minutes and the cashier can serve 10 customers in 5 minutes. Assuming Poisson arrival rate and exponential distribution for service rate, find-
 - (i) average number of customers in the queue and in the system.
 - (ii) Average time a customer spends in the queue and in the system.
 - (iii) Probability that the waiting time of a customer in the system exceeds 6 mins and the probability that the number of customers in the system exceeds 5.
3. Patients arrive at an hospital for emergency service at the rate of 2 every hour. Currently only one emergency can be handled at a time. patients spend an average 20 min. for receiving emergency service. Find-
 - (a) the average number of patients waiting in for the service and in the hospital.
 - (b) how much the average service time need to be decreased to keep the average time to wait and receive service less than 25 min?
4. Customers arrive at a one man barber shop according to a Poisson process with a mean inter-arrival time of 12 min.. Customers spend an average 10 min. in barber's chair.
 - (a) What is the expected number of customers in the shop and in the queue?
 - (b) Calculate the percentage of time an arrival can walk straight into the barber's chair without having to wait.
 - (c) How much time can a customer expect to spend in the barber's shop?
 - (d) Management will provide another chair and hire another barber, when a customer's waiting time in the shop exceeds 1.25 h. How much must be the average rate of arrivals increase to warrant a second barber?
 - (e) What is the average time customers spend in the queue?
 - (f) What is the probability that the waiting time in the system is greater than 30 min.?
 - (g) Calculate the percentage of the customers who have to wait prior to getting into barbers chair.
 - (h) What is the probability that more than 3 customers are in the system?
5. Assume that there is only one counter in a super market at which the customer arrive in a Poisson fashion at an average rate of 15 per hour and that the length of the service by the clerk has an exponential distribution. At what average must the clerk work in order to ensure a probability of 0.90 that the customer will not wait longer than 12 min.?

6. The mean rate of arrival of planes at an airport during the peak period is 20 per hour, but the actual number of arrivals in any hour follows a Poisson distribution. The airport can land 60 planes per hour on an average in good weather or 30 planes per hour in bad weather but the actual number landed in any hour follows a Poisson distribution with respective averages. When there is congestion, the planes are forced to fly over the field in the stack awaiting the landing of other planes that arrived earlier.
- (a) How many planes would be flying over the field in the stack on an average in good weather and in bad weather?
- (b) How long a plane would be in the stack and in the process of landing in good and bad weathers?
- (c) How much stack and landing time to allow so that priority to land out of order will have to be requested only 1 in 20 times?

Answers

1. (i) $\frac{25}{24}$ (ii) $\frac{5}{3}$ (iii) $\frac{5}{6}$ hours (iv) $\frac{4}{3}$ (v) 0.41
2. (i) 9 (ii) 5 min (iii) 0.53
3. (i) 2 (ii) 6.36 mins
4. (a) 4 (b) $\frac{1}{6}$ (c) 60 min. (d) $\frac{13}{6}$ (e) 50 min (f) 0.61 (g) $\frac{5}{6}$ (h) 0.48
5. $\mu = 24/h$
6. (a) $E(N_q) = \frac{\lambda^2}{\mu(\mu - \lambda)} = \begin{cases} 1/6, & \text{in good weather} \\ 4/3, & \text{in bad weather} \end{cases}$
- (b) $E(W_s) = \frac{1}{(\mu - \lambda)} = \begin{cases} 1/40 h, & \text{in good weather} \\ 1/10 h, & \text{in bad weather} \end{cases}$
- (c)

$$\begin{aligned}
 P(W_s > t_r) &= 0.05 \\
 \implies &\begin{cases} e^{-40t_r} = 0.05, & \text{in good weather} \\ e^{-10t_r} = 0.05, & \text{in bad weather} \end{cases} \\
 \implies &\begin{cases} t_r = 0.075h, & \text{in good weather} \\ t_r = 0.299h, & \text{in bad weather} \end{cases}
 \end{aligned}$$

Exercise 7.2 *Infinite capacity, multi-server Poisson queue model*

1. A tax-consulting firm has 3 counters in its office to receive people who have problems concerning their income, wealth and sales taxes. On the average 48 persons arrive in an 8-hour day. Each tax adviser spends 15 minutes on the average on an arrival. If the arrivals are Poisson distributed and service times are according to exponential distribution, find-
 - (i) the probability that a customer has to wait before he gets service
 - (ii) Average time a customer has to spend in the queue and in the system.
2. A telephone exchange has two long distance operators. The telephone company finds that during the peak load, long distance calls arrive in a Poisson fashion at an average rate of 15 per hour. The length of the service on these calls is approximately exponentially distributed with mean length of 5 minutes.
 - (a) What is the probability that a subscriber has to wait for his long distance call during the peak hours of the day?
 - (b) If the subscriber will wait and serviced in turn, what is the expected waiting time?
3. There are three typists in an office. Each typist can type an average of 6 letters per hour. If letters arrive for being typed at the rate of 15 per hour,
 - (a) What fraction of time will all the typists be busy?
 - (b) What is the average time a letter has to spend for waiting and for being typed?
4. A dress shop has three sales persons. Assume that arrivals follow Poisson pattern with an average of 10 min between arrivals. Also assume that any sales person can provide the desired service for any customer. If the time to provide service for customer is exponentially distributed with mean of 20 min per customer, calculate $E(N_s)$, $E(N_q)$, $E(W_s)$, $E(W_q)$ and P_n for $n = 0, 1, 2$.
5. If the mean arrival rate is $24/h$, find from the customer's point of view of the time spent in the system, whether 3 channels in parallel with a mean service rate of 10 per hour is better or worse than a single channel with mean service rate of $30/h$.

Answers

1. (i) 0.24 (ii) 0.29h and 0.4h.
2. (a) 0.47 (b) 3.1 mins.
3. (a) 0.70 (b) 24 mins.
4. 2.89, 0.89, 28.889 mins, 8.89 mins, 1/9, 2/9, 2/9
5. Single channel is better.

Exercise 7.3 *Finite capacity single server Poisson queue Model (M/M/1):(k/FIFO)*

1. The local one-person barber shop can accommodate a maximum of 5 people at a time (4 waiting and 1 getting hair-cut). Customer arrive according to a Poisson distribution with mean 5 per hour. The barber cuts hair at an average rate of 4 per hour (exponential service time).
 - (a) What percentage of time is the barber idle?
 - (b) What fraction of potential customers are turned away?
 - (c) What is the expected number of customers waiting for a hair-cut?
 - (d) How much time can a customer expect to spend in the barber shop?
2. In a single server Queuing system with Poisson input and exponential service times, if the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hour and the maximum possible number of calling units in the system is 3, find-
 - (i) the average number of calling units in the queue.
 - (ii) the average time spent in the queue and in the system.
3. At a railway station, only one train is handled at a time. The railway yard is sufficient for only 2 trains to wait, while the other is given signal to leave the station. Trains arrive at the station at an average rate of 6 per hour and the railway station can handle them on an average of 12 per hour. Assume Poisson arrivals and exponential service distribution, find-
 - (i) the average number of trains in the queue.
 - (ii) the average time spent by a train in the queue and in the system.

Answers

1. (a) 0.09 (b) 0.27 (c) 2.2 (d) 0.86h
2. (i) 0.51 (ii) 0.20
3. (i) 0.26 (ii) 0.013h, 0.05h

Exercise 7.4 *Finite capacity multiple server Poisson queue Model (M/M/s):(k/FIFO)*

1. A small mail-order business has one telephone line and a facility for call waiting for two additional customers. Orders arrive at the rate of one per minute and each order requires 2 minutes and 30 seconds to take down the particulars.
 - (i) What is the expected number of calls waiting in the queue? What is the mean wait in queue?
 - (ii) Because of the excessive waiting time, the business decides to use two telephone lines instead of one, keeping the same total capacity for the number in the system, namely 3. What improvements result in the performance measures considered under (i)?
 - (iii) What is the impact of increasing the capacity to four customers in the system in part (ii)?
2. A group of engineers has 2 terminals available to aid their calculations. The average computing job requires 20 min of terminal time each engineer requires some computation about once every half an hour. Assume that these are distributed according to an exponential distribution. If there are 6 engineers in the group, find-
 - (i) the expected number of engineers waiting to use one of the terminals and in the computing center.
 - (ii) the total time lost per day.
3. At a port there are 6 unloading berths and 4 unloading crews. When all the berths are full, the arriving tankers are diverted to an overflow facility 20 kms down the river. Tankers arrive according to a Poisson process with a mean of 1 every 2 hour. It takes for an unloading crew, on an average 10h to unload a tanker, the unloading time following an exponential distribution. Find
 - (a) How many tankers are at the port on the average?
 - (b) How long does a tanker spend at the port on the average?
 - (c) What is the average arrival rate at the overflow facility?
4. An automobile inspection station has 3 inspection stalls. Assume that cars wait in such a way that when a stall becomes vacant, the car at the head of the line pulls upto it. The station can accommodate at most 4 cars waiting at one time. The arrival pattern is Poisson with a mean of 1 car every minute during the peak hours. The service time is exponential with mean 6 minutes. Find the average number of cars in the system during peak hours, the average waiting time and the average number of cars per hour that can not enter the station because of full capacity.

Answers

1. (i) 3.85min (ii) 1.10min (iii) 1.64, 5.62min.
2. (i) 0.74 (ii) 3.82h
3. (a) 4.35 (b) 12.32 (c) 0.15
4. 3.18, 6.14, 12.41min, 31.10

Exercise 8.1 *Test of significance of the difference between sample proportion and population proportion.*

1. A candidate at an election claims 90% of support of all voters in a locality. Verify his claim; if in a random sample of 400 voters from the locality, 320 supported his candidature. Determine the confidence interval for the true proportion of voters supporting the candidate at a 5% level of significance.
2. A candidate for an election from a large constituency thinks that he will win the election if at least 45% of the electorate vote for him. He, therefore, conducts a sample survey to enable him to decide whether he should stand for the election or not. The survey covers 10000 voters and it is found that 4420 voters voted for him. Advise him as to whether he should stand for the election based on testing the hypothesis at 5% level of significance.
3. Experience has shown that 20% of a manufactured product is of top quality. In one day's production of 400 articles, only 50 are of top quality. Test whether the production of the day chosen was not a representative sample or the hypothesis of 20% was wrong. Also find the 99% confidence limits for the percentage of top quality.
4. A salesman in a departmental store claims that at most 60% of the shoppers entering in the store leave without making a purchase. A random sample of 50 shoppers showed that 36 of them left without making a purchase. Are these sample results consistent with the claim of the salesman? If not, find the 95% confidence limits for P .
5. The fatality rate of a virus is believed to be 18.36%. In a certain year 450 patients suffering from the virus were treated with a vaccine and only 68 patients died. Can the vaccine be considered as efficient? [LOS 2%]
6. A manufacturer of lenses is qualifying a new grinding machine and will qualify the machine if the percentage of polished lenses that contain surface defects does not exceed 2.25%. A random sample of 250 lenses contains 10 defective lenses. Test at 5% level of significance if the machine can be qualified.

Answers

1. (0.7608, 0.8392)
2. He should stand for the election.
3. The hypothesis of 20% was wrong, (0.0823, 0.1677)
4. The salesman's claim is not true, the salesman's claim must be atleast 61.56%
5. The vaccine is not efficient.
6. The machine does not qualify.

Exercise 8.2 *Test of the significance of difference between two sample proportions.*

1. In a test of the quality of two television commercials, each commercial was shown in a separate test area six times over a one-week period. The following week a telephonic survey was conducted to identify individuals who had seen the commercials. Those individuals were asked to state the primary message in the commercials. The following results were recorded:

	Commercial A	Commercial B
Number who saw commercial	150	200
Number who recalled message	63	60

- (a) Use $\alpha = 0.05$ and test the hypothesis that there is no difference in the recall proportions for the two commercials.
 - (b) Compute a 95% confidence interval for the difference between the recall proportions for the two populations.
2. A machine produced 20 defective articles in a batch of 400. After overhauling, it produced 10 defectives in a batch of 300. Has the machine improved? Test the hypothesis at 5% level of significance.
 3. Researchers conducted a study of smart phone use among adults. A cell phone company claimed that iPhone smart phones are more popular with female than with male. The results of the survey indicate that of the 1343 male cell phone owners randomly sampled, 337 own an iPhone while of the 232 female cell phone owners randomly sampled, 52 have an iPhone. Is the proportion of male iPhone owners greater than the proportion of female iPhone owners? Test at the 5% level of significance. Also find the 95% confidence intervals for the difference in population proportions of male and female iPhone owners.
 4. In a large city A, 20% of a random sample of 700 school boys had a slight physical defect. In another large city B, 18.5% of a random sample of 1200 school boys had same defect. Is the difference between the proportions significant? [LOS 5%].
 5. 16.35% of a random sample of 1600 undergraduates were smokers, whereas 20% of a random sample of 900 postgraduates were smokers in a state. Can we conclude that less number of undergraduates are smokers than the postgraduates? [LOS 1%].
 6. A drug company develops a new drug, designed to prevent colds. The company states that the drug is equally effective for men and women. To test this claim, they choose a simple random sample of 100 women and 200 men from a population of 100,000 volunteers. At the end of the study, 38% of the women caught a cold; and 51% of the men caught a cold. Based on these findings, can we reject the company's claim and conclude that the drug is more effective for women? Use a 0.02 level of significance.

Answers

1. (a) There is significant difference in the recall proportions for the two commercials (b) (0.3, 0.4)
2. The machine has not improved.
3. The proportions are the same, $-0.02 \leq P_1 - P_2$
4. The difference not significant.
5. We can not conclude that less number of undergraduates are smokers than the post graduates.

6. The drug is more effective for women.

Exercise 8.3 *Test of the significance of difference between sample mean and population mean.*

1. A weighing machine without any display was used by an average of 260 persons a day with a standard deviation of 45 persons. When an attractive display was used on the machine, the average for 150 days increased by 20 persons. Can we say that the display did not help much? Use a level of significance of 0.05.
2. A manufacturer claims that the mean breaking strength of safety belts for air passengers produced in his factory is 1275 kg. A sample of 125 belts was tested and the mean breaking strength and *SD* were found to be 1260 kg and 90 kg respectively. Test the manufacturer's claim at 5% LOS.
3. An *IQ* test was given to a large group of boys in the age group of 18 – 20 years, who scored an average of 61.5 marks. The same test was given to a fresh group of 100 boys of the same age group. They scored an average of 64.5 marks with an *SD* of 12.5 marks. Can we conclude that the fresh group of boys have better *IQ*? Test at 2% LOS. Also find the 98% confidence interval for the average marks.
4. The guaranteed average life of a certain brand of electric bulb is 1025 hours with a *SD* of 125 hours. In a sample of 200 bulbs the average life is found to be 990 hours. Can we conclude that the average life of bulbs is not as guaranteed? Also find the 99% confidence interval for the average life of the bulbs.

Answers

1. Yes, helpful.
2. The claim is true.
3. The fresh group of boys have better *IQ*. The confidence interval is (61.931, 64.5)
4. The average life of bulbs is less than 1025 hours. The confidence interval is (990, 1010.6)

Exercise 8.4 *Test of the significance of difference between two sample means.*

1. A brochure inviting subscriptions for a new diet program states that the participants are expected to lose over 22 pounds in five weeks. Suppose that, from the data of the five-week weight losses of 56 male participants, the sample mean and sample standard deviation are found to be 24.25 and 10.2 respectively and that of 50 females participants to be 20.25 and 9.7 respectively. Can it be concluded that the program is more effective for male. Also find the 98% confidence interval for the difference for the population means.
2. In a certain factory, there are two independent processes manufacturing the same item. The average weight in a sample of 250 items produced from one process is found to be 120 gms with a standard deviation of 12 gms, while the corresponding figures in a sample of 400 items from the other process are 124 gms and 14 gms. Obtain the standard error of difference between the two sample means. Is the difference significant? Also find the 99% confidence interval for difference in average weights of the items produced by the two processes respectively.
3. The Municipal Transit Authority wants to know if, on weekdays, more passengers ride the northbound blue line train towards the city center that departs at 8 : 15a.m. or the one that departs at 8 : 30a.m. The following sample statistics are assembled by the Transit Authority.

	n	$\bar{x}$	s
8 : 15am train	32	323	41
8 : 30am train	38	356	45

- (i) Construct the 98% confidence interval for the difference in the mean number of daily travelers on the 8 : 15am train and the mean number of daily travelers on the 8 : 30am train.
 - (ii) Test at the 5% level of significance whether the data provide sufficient evidence to conclude that more passengers ride the 8 : 30am train.
4. The company has developed a new battery. The engineer in charge claims that the new battery will have better performance than that of the old battery. To test the claim, the company selects a simple random sample of 90 old batteries and 65 new batteries. The old batteries ran continuously for 190 minutes with a standard deviation of 32 minutes; the new batteries, 200 minutes with a standard deviation of 45 minutes. Test the engineer's claim at 0.05 level of significance.
 5. A sample of 100 bulbs of brand A gave a mean lifetime of 1200 h with a SD of 70 h, while another sample of 120 bulbs of brand B gave a mean lifetime of 1175 h with a SD of 85 h. Can we conclude that brand A bulbs are superior to brand B bulbs (LOS 2%)? Also find the 98% confidence interval for difference between the mean lifetimes of the two brands of bulbs.
 6. The research unit in an organization wishes to determine whether scores on the scholastic aptitude test are different for male and female applicants. Random samples of applicants files are taken and summarized below-

	Applicants	
	Male	Female
Mean	502.1	515.2
SD	86.2	90.4
n	399	204

Based on the above data, can it be concluded that the average score is same for both types of applicants. Also find the confidence interval for the difference of the population means. [$\alpha = 0.05$]

7. A supermarket chain is considering two sources A and B for the purchase of 50–pound bags of onions. The following table gives the results of a study

	Source A	Source B
Number of bags weighed	60	75
Mean weight	105.9	100.5
Sample SD	10	12

Test at $\alpha = 0.05$ whether there is a difference in the mean weights. Also find the 95% confidence interval for the difference of means of weights of the bags from the two sources.

Answers

1. More effective for male. The 98% confidence interval is (0.5058, 8.5058).
2. The difference between the average weights is significant. (1.3362, 6.6639).
3. (i) (−56.969, −9.0313). (ii) More passengers ride on the 8 : 30am train.
4. The engineer's claim is wrong.
5. Brand A bulbs are superior to brand B bulbs. (0.65, 49.35)
6. The average score is the same for both types of applicants. (−33.11, 3.086)
7. The mean weights are not the same. (1.688, 9.111)

Exercise 8.5 *Sample standard deviation and population standard deviation*

1. In a certain random sample of 72 items, the *sd* is found to be 8. Is it reasonable to suppose that it has been drawn from a population with *sd* 6.5. If not find the 98% confidence interval for the *sd* of the population.
2. A manufacturer of electric bulbs, according to a certain process, finds the *sd* of the life lamps to be 100 *h*. He wants to change the process, if the new process results in a smaller variation in the life of lamps. In adopting a new process, a sample of 150 bulbs gave an *sd* of 95 *h*. Is the manufacturer justified in changing the process?{LOS 5%}
3. In a random sample of 200 items, drawn from a population with *sd* 0.8, the sample *sd* is 0.9. Can we conclude that the population *sd* is less than the sample *sd*? If yes, find the 99% confidence interval for the population *sd* to be regarded same as the sample *sd*.
4. The *sd* of a random sample of 1000 is found to be 2.6 and the *sd* of another random sample of 500 is 2.7. Assuming that the samples to be independent, find whether the two samples could have come from populations with same *sd*.

Answers

1. The random sample has not been drawn from a population with *sd* 6.5. (6.73,9.26)
2. The manufacturer is not justified in changing the process.
3. Yes, the population *sd* is less than the sample *sd*. $\sigma \geq 0.81$.
4. Yes, the two samples could have come from populations with same *sd*.

8.10.4 Critical values of t and the t -table.

The critical values $t_v(\alpha)$ of t at $LOS \alpha\%$ for various degrees of freedoms v have been tabulated in a table for two-tailed test by R.A. Fisher in a table called the t – table. The critical value $t_v(\alpha)$ for a one-tailed test (right or left) at $LOS \alpha$ is same as $t_v(2\alpha)$ for a two-tailed test.

Level of Significance for two tailed test					
	0.1	0.05	0.04	0.02	0.01
Level of Significance for one tailed test					
df	0.05	0.025	0.02	0.01	0.005
1	6.3138	12.7062	15.8945	31.8205	63.6567
2	2.9200	4.3027	4.8487	6.9646	9.9248
3	2.3534	3.1824	3.4819	4.5407	5.8409
4	2.1318	2.7764	2.9985	3.7469	4.6041
5	2.0150	2.5706	2.7565	3.3649	4.0321
6	1.9432	2.4469	2.6122	3.1427	3.7074
7	1.8946	2.3646	2.5168	2.9980	3.4995
8	1.8595	2.3060	2.4490	2.8965	3.3554
9	1.8331	2.2622	2.3984	2.8214	3.2498
10	1.8125	2.2281	2.3593	2.7638	3.1693
11	1.7959	2.2010	2.3281	2.7181	3.1058
12	1.7823	2.1788	2.3027	2.6810	3.0545
13	1.7709	2.1604	2.2816	2.6503	3.0123
14	1.7613	2.1448	2.2638	2.6245	2.9768
15	1.7531	2.1314	2.2485	2.6025	2.9467
16	1.7459	2.1199	2.2354	2.5835	2.9208
17	1.7396	2.1098	2.2238	2.5669	2.8982
18	1.7341	2.1009	2.2137	2.5524	2.8784
19	1.7291	2.0930	2.2047	2.5395	2.8609
20	1.7247	2.0860	2.1967	2.5280	2.8453
21	1.7207	2.0796	2.1894	2.5176	2.8314
22	1.7171	2.0739	2.1829	2.5083	2.8188
23	1.7139	2.0687	2.1770	2.4999	2.8073
24	1.7109	2.0639	2.1715	2.4922	2.7969
25	1.7081	2.0595	2.1666	2.4851	2.7874
26	1.7056	2.0555	2.1620	2.4786	2.7787
27	1.7033	2.0518	2.1578	2.4727	2.7707
28	1.7011	2.0484	2.1539	2.4671	2.7633
29	1.6991	2.0452	2.1503	2.4620	2.7564

Exercise 8.6 *Test of significance of difference between Sample mean and population mean.*

1. A pharmaceutical company claims that their new drug reduces blood pressure by an average of 8 mmHg. A group of researchers tests the drug on 25 patients and finds that the average reduction in blood pressure is 7 mmHg with a standard deviation of 1.5 mmHg. Use a significance level of 0.05 to test whether the drug's effectiveness is as claimed. ($t = 2.064$).
2. A random sample of 8 envelopes is taken from the letter box of a post office and their weights in grams are found to be 12.2, 11.9, 12.5, 12.3, 11.6, 11.7, 12.2 and 12.4. Can it be concluded that the mean weight of all the envelopes in the letter box is 11.7 grams. Also find the 98% confidence limits for the mean weight of the envelopes in the letter box.
3. The annual rainfall at a certain place is normally distributed with mean 30. If the rainfalls during the past 8 years are 31.1, 30.7, 24.3, 28.1, 27.9, 32.2, 25.4 and 29.1, can it be concluded that the average rainfall during the past 8 years is less than the normal rainfall? [$\alpha = 0.02$]
4. The mean weekly sales of soap bars in departmental stores is 145. After an advertising campaign, the mean weekly sales in 17 stores for a typical week increased to 155 and showed an *SD* of 16. Was the advertising campaign successful? [$\alpha = 0.05$]
5. A random sample of 16 values from a normal population showed a mean of 103.75cm and the sum of squares of the deviations from the mean is 843.75cm^2 . Show that the assumption of a mean of 108.75cm for the population is not reasonable. Obtain 95% and 99% fiducial limits for the same.
6. A sample of 9 shafts is inspected from a production line. The following measurements are the diameters (in mm) of shafts:

45.010, 45.020, 45.021, 45.015, 45.019, 45.018, 45.020, 45.023, 45.005

If the production line meets the specifications laid by the *I.S.I.*, estimate the 95% and 98% confidence intervals within which the true diameter of the shaft lies.

Answers

1. The average reduction might be less than 8 mmHg.
2. The mean weight of the letters in the letter box is not 11.7 grams. (11.74, 12.46)
3. The average rainfall during the past 8 years is same as the normal rainfall.
4. The advertising campaign was successful.
5. The assumption of a mean of 108.75cm for the population is not reasonable.
(99.63, 107.87), (98.04, 109.46)
6. (45.013, 45.021), (45.011, 45.023)

Exercise 8.7 *Test of significance of difference between means of two independent small samples.*

1. To compare two kinds of front-end designs, six of each kind was installed on a certain make of compact car. Then each car was run into a concrete wall at 5 miles per hour, and the following are the costs of the repairs (in dollars):

Design 1:	127	168	143	165	122	139
Design 2:	154	135	132	171	153	149

Test the hypothesis at the 0.01 level of significance whether the difference between the means of these two samples is significant. [$t_{0.01} = 3.169$].

2. The College Board provided comparisons of Scholastic Aptitude Test (SAT) scores based on the highest level of education attained by the test takers parents. A research hypothesis was that students whose parents attained a higher level of education would on average score higher on the SAT. SAT verbal scores of independent samples of students (based on their parents education) are given as follows. Test the hypothesis at 5% level of significance. [$t_{0.05} = 1.761$]

Bachelors Degree	485	650	550	534	554	592	572	578
High school degree	442	580	528	425	478	486	524	535

3. Vaccines *A* and *B* were administered simultaneously to two sets *A* and *B* of people to test any side effects. A random sample of 12 people from the set *A* and a random sample of 14 people from set *B* were chosen and the weight loss in them after a certain period of time were collected as given below:

<i>A</i>	5	7	9	4	8	6	2	3	7	6	4	8		
<i>B</i>	3	8	5	7	2	3	6	4	9	4	5	5	7	2

Test at 1% level of significance that the vaccine *B* is safe.

4. To compare two kinds of bumper guards, each kind were mounted on a certain make of compact car. Then each car was run into a concrete wall at 5 miles per hour, and the following are the costs of the repairs (in dollars):

Bumper guard 1	127	168	143	165	122	139	
Bumper guard 2	154	135	132	171	153	149	172

Test, at the $\alpha = 0.01$ significance level, whether the cost of repairs for the bumper guard 1 is less than that of the bumper guard 2.

5. A research team wants to investigate the usefulness of relaxation training for reducing levels of anxiety in individuals experiencing stress. They identify 15 people at random from a group of 100 who have "high stress" jobs. The 15 people are divided into two groups. One group of 7 called "control group" receives no training. The second group of 8 called "relaxation group" receives the relaxation training. The subjects in each group are then given an anxiety test. The scores are given in the following table where higher scores indicate greater anxiety-

Control	38	40	35	36	35	32	31
Relaxation	35	32	30	34	30	32	24

Test whether the relaxation training has reduced the anxiety. [$\alpha = 0.01$]

6. A random sample of 8 envelopes is taken from the letter box of a post office and their weights in grams are found to be 12.2, 11.9, 12.5, 12.3, 11.6, 11.7, 12.2 and 12.4. Can it be concluded that the mean weight of all the envelopes in the letter box is 11.7 grams. Also find the 98% confidence limits for the mean weight of the envelopes in the letter box.
7. Two different types of drugs *A* and *B* were tried on certain patients for increasing weight. Five persons were given drug *A* and 7 persons were given the drug *B*. The increase in weight (in kgs) is given below-

Drug <i>A</i>	3.6	5.5	5.9	4.1	1.4				
Drug <i>B</i>	4.5	3.6	5.5	6.8	2.7	3.6	5.0		

Can it be concluded at $\alpha = 0.05$ that the drug *B* is more effective than drug *A* in increasing weight?

8. The following data shows the cost in hundreds rupees per square meter of the floor area concerning randomly selected 7 schools and 5 office blocks from those completed during a certain period-

Schools	28	31	26	27	23	38	37
Office blocks	37	42	34	37	35		

Do the data support the hypothesis that the cost per square meter for the office blocks was greater than that for the schools.

9. Two independent groups of 10 children were tested to find how many digits they could repeat from memory after hearing. The results are as follows-

Group <i>A</i>	8	6	5	7	6	8	7	4	5	6
Group <i>B</i>	10	6	7	8	6	9	7	6	7	7

Is the difference between mean scores of the two groups significant at *LOS* 1%.

Answers

1. The mean repair costs for the two designs are the same.
2. The hypothesis is wrong.
3. The two vaccine have the same side effects.
4. The cost of repairs for the bumper guard 1 and that of the bumper guard 2 are the same.
5. The relaxation training has not reduced the anxiety.
6. No. (11.74, 12.46)
7. No.
8. No.
9. No.

Exercise 8.8 *Test of two dependent (correlated) samples*

1. Ten soldiers visit a rifle range for two consecutive weeks. For the first week their scores are: 67, 24, 57, 55, 63, 54, 56, 68, 33, 43 and during the second week their score in the same order: 70, 38, 58, 58, 56, 67, 68, 72, 42, 38. Examine if there is any significant improvement in the performance. ($t_{0.05} = 1.833$)
2. Engineers at a tire manufacturing corporation wish to test a new tire material for increased durability. To test the tires under realistic road conditions, new front tires are mounted on each of 11 company cars, one tire made with a production material and the other with the experimental material. After a fixed period the 11 pairs were measured for wear. The amount of wear for each tire (in mm) is shown in the table-

Car	1	2	3	4	5	6	7	8	9	10	11
Production Material	5.1	6.5	3.6	3.5	5.7	5.0	6.4	4.7	3.2	3.5	6.4
Experimental Material	5	6.5	3.1	3.7	4.5	4.1	5.3	2.6	3	3.5	5.1

- (i) Construct the 99% confidence interval for the difference based on these data.
 - (ii) Test, at the 1% level of significance, the hypothesis that the mean wear with the experimental material is less than that for the production material.
3. An intensive coaching was given to 11 students and they were examined 5 times in a month. The results of first and fifth tests are given below-

Sr. No	1	2	3	4	5	6	7	8	9	10	11
Test-I	50	42	51	26	35	42	60	41	70	55	62
Test-V	62	40	61	35	30	52	68	51	84	63	72

State whether there has been an improvement in the result of fifth test over the first test? Test at 5% level of significance. [$t_{0.05} = 1.182$]

4. A Wire cable is manufactured by two processes. Laboratory tests were performed by putting samples of cables under tension and recording the load required (coded units) to break the cable, giving the following data:

Process 1	9	4	10	7	9	10
Process 2	14	9	13	12	13	8

Can we say that the two processes have same effect on the mean breaking strength, at 5% level of significance? ($t_{0.05} = 2.228$).

5. The following data relate to the marks obtained by 11 students in 2 tests, one held at the beginning of a year and the other at the end of the year after intensive coaching-

Test I	19	23	16	24	17	18	20	18	21	19	20
Test II	17	24	20	24	20	22	20	20	18	22	19

6. A new diet and exercise program has been advertised as remarkable way to reduce blood glucose levels in diabetic patients. Ten randomly selected diabetic patients are put on the program, and the results after 1 month are given by the following table-

Before	268	225	252	192	307	228	246	298	231	185
After	106	186	223	210	203	101	211	176	194	203

Do the data provide sufficient evidence at $\alpha = 0.01$ to support the claim that the new program reduces blood glucose level in diabetic patients? [$|t_{0.01}(9)| = 2.821$]

Answers

1. Significant improvement in the performance of the soldiers in the second week.
2. (i) (0.0675, 1.3765) (ii) The mean wear with the experimental material is less than that for the production material.
3. Significant improvement in the result of fifth test over the first test.
4. The two processes differ significantly as regards to the effect on mean breaking strength.
5. The students are not benefited by the coaching.
6. The new program reduces blood glucose level in diabetic patients.

	χ^2 table							
	LOS $\alpha\%$							
ν	0.1	0.05	0.025	0.02	0.01	0.005	0.002	0.001
1	2.706	3.841	5.024	5.412	6.635	7.879	9.55	10.828
2	4.605	5.991	7.378	7.824	9.21	10.597	12.429	13.816
3	6.251	7.815	9.348	9.837	11.345	12.838	14.796	16.266
4	7.779	9.488	11.143	11.668	13.277	14.86	16.924	18.467
5	9.236	11.07	12.833	13.388	15.086	16.75	18.907	20.515
6	10.645	12.592	14.449	15.033	16.812	18.548	20.791	22.458
7	12.017	14.067	16.013	16.622	18.475	20.278	22.601	24.322
8	13.362	15.507	17.535	18.168	20.09	21.955	24.352	26.124
9	14.684	16.919	19.023	19.679	21.666	23.589	26.056	27.877
10	15.987	18.307	20.483	21.161	23.209	25.188	27.722	29.588
11	17.275	19.675	21.92	22.618	24.725	26.757	29.354	31.264
12	18.549	21.026	23.337	24.054	26.217	28.3	30.957	32.909
13	19.812	22.362	24.736	25.472	27.688	29.819	32.535	34.528
14	21.064	23.685	26.119	26.873	29.141	31.319	34.091	36.123
15	22.307	24.996	27.488	28.259	30.578	32.801	35.628	37.697
16	23.542	26.296	28.845	29.633	32	34.267	37.146	39.252
17	24.769	27.587	30.191	30.995	33.409	35.718	38.648	40.79
18	25.989	28.869	31.526	32.346	34.805	37.156	40.136	42.312
19	27.204	30.144	32.852	33.687	36.191	38.582	41.61	43.82
20	28.412	31.41	34.17	35.02	37.566	39.997	43.072	45.315
21	29.615	32.671	35.479	36.343	38.932	41.401	44.522	46.797
22	30.813	33.924	36.781	37.659	40.289	42.796	45.962	48.268
23	32.007	35.172	38.076	38.968	41.638	44.181	47.391	49.728
24	33.196	36.415	39.364	40.27	42.98	45.559	48.812	51.179
25	34.382	37.652	40.646	41.566	44.314	46.928	50.223	52.62
26	35.563	38.885	41.923	42.856	45.642	48.29	51.627	54.052
27	36.741	40.113	43.195	44.14	46.963	49.645	53.023	55.476
28	37.916	41.337	44.461	45.419	48.278	50.993	54.411	56.892
29	39.087	42.557	45.722	46.693	49.588	52.336	55.792	58.301
30	40.256	43.773	46.979	47.962	50.892	53.672	57.167	59.7

Exercise 8.9 *Test of goodness of fit*

1. In an industry, 200 workers, employed for a specific job, were classified according to their performance and training received/ not received to test independence of a specific training and performance. The data summarized as follows:

	Performance		Total
	Good	Not good	
Trained	100	50	150
Untrained	20	30	50
Total	120	80	200

Use chi-square test of independence at $\alpha = 0.05$ and write your conclusion. [$\chi_{0.05}^2 = 3.841$].

2. The demand for a particular spare part in a factory was found to vary from day to day. In a sample study the following information was obtained-

Days	Mon	Tue	Wed	Thu	Fri	Sat
No of parts demanded	1124	1125	1110	1120	1126	1115

Test the hypothesis at 5% level of significance that the number of parts demanded does not depend on the day of the week.

3. Theory predicts that the proportion of beans in 4 groups A, B, C and D should be $9 : 3 : 3 : 1$. In an experiment among 1600 beans, the numbers in the 4 groups were 872, 323, 277 and 128. Does experiment support the theory?
4. The following data give the number of air-craft accidents that occurred during the various days of a week-

Day	Mon	Tue	Wed	Thu	Fri	Sat
No. of Accidents	15	19	13	12	16	15

Test whether the accidents are uniformly distributed over the week. [$\alpha = 1\%$].

5. Genetic theory states that the children having one parent of blood type M and other blood type N will always be one of the three types M, MN and N and that the portion of these types will on average be $1 : 2 : 1$. A report states that out of 300 children having one M parent and one N parent, 30% were found to be of type M , 45% of type MN and rest of type N . Test the hypothesis at 5%.
6. Fit a binomial distribution to the data-

x	0	1	2	3	4	5
$f(x)$	38	144	342	287	164	25

and test the goodness of fit at LOS 5%.

7. The frequencies of localities according to the number of deaths due to cholera during certain period in 1500 localities is as follows-

Deaths	0	1	2	3	4	5	6	7
Localities	814	378	148	79	67	9	3	2

Fit a Poisson distribution to the data and test the goodness of fit at LOS 2%.

8. Obtain the equation of the normal curve that may be fitted to the data and test the goodness of fit at LOS 2%.

x	4	6	8	10	12	14	16	18	20	22	24
$f(x)$	1	7	15	22	35	43	38	20	13	5	1

Answers

1. Performances are associated.
2. The number of parts demanded does not depend on the day of the week.
3. The experiment does not support the theory.
4. The accidents are uniformly distributed over the week.
5. The binomial distribution does not fit good to the given data.
6. The Poisson distribution fits good to the given data.

Exercise 8.10 *Test of independence of attributes*

1. Genetic theory states that children having one parent of blood type A and the other of blood type B will always be of one of three types, A, AB, B and that the proportion of three types will on an average be as $1 : 2 : 1$. A report states that out of 300 children having one A parent and B parent, 30 percent were found to be types A , 45 percent type AB and remainder type B . Test the hypothesis at the 0.05 level of significance. [$\chi_{0.05}^2 = 5.991$]
2. It is generally believed that children brought up in stable families tend to do well in school. To verify such a belief, a social scientist examined 290 randomly selected students records in a public high school and noted each students family structure and academic status four years after entering high school. The data were then sorted into a 3×3 contingency table with two factors. Factor 1 has three levels: school, graduate and postgraduate. Factor 2 has three levels: no parent, one parent, and two parents. The results are given in the table provided.

		Academic Status		
		School	Graduate	Post graduate
Family	No parent	15	32	35
	One parent	92	52	48
	Two parents	48	22	25

Test, at the 5% level of significance, whether there is sufficient evidence in the data to conclude that family structure matters in school performance of the students.

3. A lending institution supplied the following data on loan approvals by four loan officers. Use 5% level of significance and determine whether the loan approval decision is independent of the loan officer reviewing the loan application. [$\chi_{0.05}(3) = 7.815$] **[CT-III IT-III Nov 2022]**

Loan officer	Loan Approval decision	
	Approved	Rejected
A	24	16
B	17	13
C	35	15
D	11	9

4. The following sample data pertains to the shipment received by a large firm from four different vendors:

Quality	Vendors			
	A	B	C	D
Rejected	30	15	15	18
Imperfect but acceptable	50	10	40	34
Perfect	90	50	60	78

Test at $\alpha = 0.05$, whether the three vendors ship products of equal quality.

5. The results of a survey regarding radio listeners preference for different types of music are given in the following table, with listeners classified by the age group. Is preference

of type of music influenced by age? [Test at LOS 5%.]

Type of music	Age group		
	19 – 25	26 – 35	Above 36
National	80	60	90
Foreign	210	325	44
classical	16	45	132

6. Test the independence of the tyre brands and the distance covered using these tyres at 2% LOS.

Distance	Tyre brands			
	A	B	C	D
Failed to last 40000 kms	26	23	15	32
lasted from 40000 – 60000 kms	118	93	116	121
lasted more than 60000 kms	56	84	69	47

7. The following table gives the opinion on collective bargaining by a random sample of 360 employees of a school system, belonging to a teachers' union.

	For	Against	Undecided
Staff	30	15	15
Faculty	50	10	40
Administration	90	50	60

For $\alpha = 0.01$, test whether the opinion on collective bargaining is independent of employee classification.

8. The following table gives a classification according to religious affiliation and marital status for 500 randomly selected individuals. For $\alpha = 0.01$, determine whether the marital status and religious affiliation are independent. [$\chi^2_{0.01}(4) = 13.28$]

Marital Status	Religious affiliation				
	A	B	C	D	None
Single	39	19	12	28	18
Married	172	61	44	70	37

9. The employees in 4 different firms are distributed in 3 groups shown in the table. Test the hypothesis that there is no relationship between the firm and the type of labor-

Type of labour		Firm			
		A	B	C	D
	Skilled	24	24	23	49
	Semi-skilled	32	60	37	51
	Manual	24	56	40	80

10. A random sample was taken of 300 undergraduate students from a university. The students in the sample were classified according to their gender and according to the choice of their major in the following table. Is the choice of the major by undergraduate students in this

university independent of their gender at $\alpha = 0.01$?

		<i>College</i>			
<i>Gender</i>		Arts	Engineering	Business	Others
	Male	75	40	24	66
	Female	45	12	15	23

11. Tests of the fidelity and the selectivity of 190 radios produced the following results:

		<i>Fidelity</i>		
<i>Selectivity</i>		Low	Average	High
	Low	7	12	21
	Average	35	59	18
	High	15	13	10

Test whether the fidelity is independent from selectivity. [$\alpha = 0.05$]

12. The results of a random sample of children with pain from musculoskeletal injuries treated with acetaminophen, ibuprofen, or codeine are shown in the table.

	Acetaminophen	Ibuprofen	Codeine
Significant Improvement	78	61	86
Slight Improvement	58	84	62
No Improvement	67	54	76

At $\alpha = 0.10$, is there enough evidence to conclude that the treatment and response are independent? [$\chi^2_4(0.10) = 7.779$].

Answers

1. No significant difference.
2. The family structure matters in school performance of the students.
3. The Loan approval decision is independent of the loan officer receiving the loan application.
4. The three vendors do not ship products of equal quality.
5. Influenced by age.
6. The tyre brands and the distance covered are independent.
7. The opinion on collective bargaining is independent of employee classification.
8. The Marital status and religious affiliation are independent.
9. There is a no relationship between the firm and the type of labor.
10. The choice is independent of their gender.
11. Fidelity is not independent from selectivity.
12. The treatment and response are not independent.

Critical Values of F-Distribution (at 5 per cent)

$\downarrow v_2 \backslash v_1 \rightarrow$	1	2	3	4	5	6	8	12	24	∞
1	161.4	199.5	215.7	224.6	230.2	234.0	238.9	243.9	249.1	243.3
2	18.51	19.00	19.16	19.25	19.30	19.33	19.37	19.41	19.45	19.50
3	10.13	9.55	9.28	9.12	9.01	8.94	8.85	8.74	8.64	8.53
4	7.71	6.94	6.59	6.39	6.26	6.16	6.04	5.91	5.77	5.63
5	6.61	5.79	5.41	5.19	5.05	4.95	4.82	4.68	4.53	4.36
6	5.99	5.14	4.76	4.53	4.39	4.28	4.15	4.00	3.84	3.67
7	5.59	4.74	4.35	4.12	3.97	3.87	3.73	3.57	3.41	3.23
8	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.28	3.12	2.93
9	5.12	4.26	3.86	3.63	3.48	3.37	3.23	3.07	2.90	2.71
10	4.96	4.10	3.71	3.48	3.33	3.22	3.07	2.91	2.74	2.54
11	4.84	3.98	3.59	3.36	3.20	3.09	2.95	2.79	2.61	2.40
12	4.75	3.88	3.49	3.26	3.11	3.00	2.85	2.69	2.51	2.30
13	4.67	3.80	3.41	3.18	3.02	2.92	2.77	2.60	2.42	2.21
14	4.60	3.74	3.34	3.11	2.96	2.85	2.70	2.53	2.35	2.13
15	4.54	3.68	3.29	3.06	2.90	2.79	2.64	2.48	2.29	2.07
16	4.49	3.63	3.24	3.01	2.85	2.74	2.59	2.42	2.24	2.01
17	4.45	3.59	3.20	2.96	2.81	2.70	2.55	2.38	2.19	1.96
18	4.41	3.55	3.16	2.93	2.77	2.66	2.51	2.34	2.15	1.92
19	4.38	3.52	3.13	2.90	2.74	2.63	2.48	2.31	2.11	1.88
20	4.35	3.49	3.10	2.87	2.71	2.60	2.45	2.28	2.08	1.84
21	4.32	3.47	3.07	2.84	2.68	2.57	2.42	2.25	2.05	1.81
22	4.30	3.44	3.05	2.82	2.66	2.55	2.40	2.23	2.03	1.78
23	4.28	3.42	3.03	2.80	2.64	2.53	2.38	2.20	2.01	1.76
24	4.26	3.40	3.01	2.78	2.62	2.51	2.36	2.18	1.98	1.73
25	4.24	3.38	2.99	2.76	2.60	2.49	2.34	2.16	1.96	1.71
26	4.22	3.37	2.98	2.74	2.59	2.47	2.32	2.15	1.95	1.69
27	4.21	3.35	2.96	2.73	2.57	2.46	2.31	2.13	1.93	1.67
28	4.20	3.34	2.95	2.71	2.56	2.45	2.29	2.12	1.91	1.65
29	4.18	3.33	2.93	2.70	2.54	2.43	2.28	2.10	1.90	1.64
30	4.17	3.32	2.92	2.69	2.53	2.42	2.27	2.09	1.89	1.62
40	4.08	3.23	2.84	2.61	2.45	2.34	2.18	2.00	1.79	1.51
60	4.00	3.15	2.76	2.52	2.37	2.25	2.10	1.92	1.70	1.39
120	3.92	3.07	2.68	2.45	2.29	2.17	2.02	1.83	1.61	1.25
∞	3.84	2.99	2.60	2.37	2.21	2.10	1.94	1.75	1.52	1.00

v_1 = Degrees of freedom for greater variance.

v_2 = Degrees of freedom for smaller variance.

Table entry for p is the critical value F^* with probability p lying to its right.

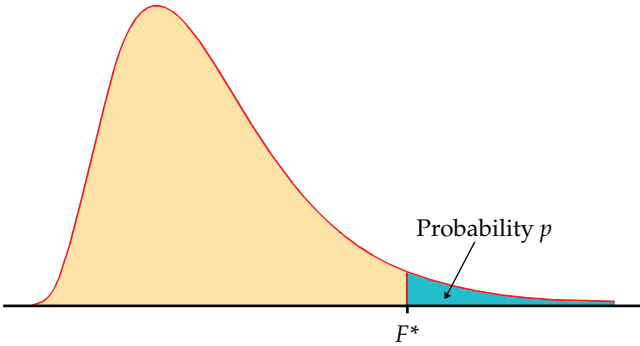
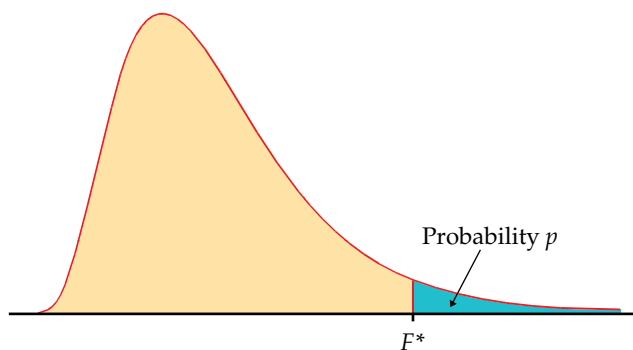


TABLE E

F critical values

		Degrees of freedom in the numerator								
p		1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	1	.100	39.86	49.50	53.59	55.83	57.24	58.20	58.91	59.44
		.050	161.45	199.50	215.71	224.58	230.16	233.99	236.77	238.88
		.025	647.79	799.50	864.16	899.58	921.85	937.11	948.22	956.66
		.010	4052.2	4999.5	5403.4	5624.6	5763.6	5859.0	5928.4	5981.1
		.001	405284	500000	540379	562500	576405	585937	592873	598144
	2	.100	8.53	9.00	9.16	9.24	9.29	9.33	9.35	9.37
		.050	18.51	19.00	19.16	19.25	19.30	19.33	19.35	19.37
		.025	38.51	39.00	39.17	39.25	39.30	39.33	39.36	39.37
		.010	98.50	99.00	99.17	99.25	99.30	99.33	99.36	99.37
		.001	998.50	999.00	999.17	999.25	999.30	999.33	999.36	999.37
	3	.100	5.54	5.46	5.39	5.34	5.31	5.28	5.27	5.25
		.050	10.13	9.55	9.28	9.12	9.01	8.94	8.89	8.85
		.025	17.44	16.04	15.44	15.10	14.88	14.73	14.62	14.54
		.010	34.12	30.82	29.46	28.71	28.24	27.91	27.67	27.49
		.001	167.03	148.50	141.11	137.10	134.58	132.85	131.58	130.62
	4	.100	4.54	4.32	4.19	4.11	4.05	4.01	3.98	3.95
		.050	7.71	6.94	6.59	6.39	6.26	6.16	6.09	6.04
		.025	12.22	10.65	9.98	9.60	9.36	9.20	9.07	8.98
		.010	21.20	18.00	16.69	15.98	15.52	15.21	14.98	14.80
		.001	74.14	61.25	56.18	53.44	51.71	50.53	49.66	49.00
	5	.100	4.06	3.78	3.62	3.52	3.45	3.40	3.37	3.34
		.050	6.61	5.79	5.41	5.19	5.05	4.95	4.88	4.82
		.025	10.01	8.43	7.76	7.39	7.15	6.98	6.85	6.76
		.010	16.26	13.27	12.06	11.39	10.97	10.67	10.46	10.29
		.001	47.18	37.12	33.20	31.09	29.75	28.83	28.16	27.65
	6	.100	3.78	3.46	3.29	3.18	3.11	3.05	3.01	2.98
		.050	5.99	5.14	4.76	4.53	4.39	4.28	4.21	4.15
		.025	8.81	7.26	6.60	6.23	5.99	5.82	5.70	5.60
		.010	13.75	10.92	9.78	9.15	8.75	8.47	8.26	8.10
		.001	35.51	27.00	23.70	21.92	20.80	20.03	19.46	19.03
	7	.100	3.59	3.26	3.07	2.96	2.88	2.83	2.78	2.75
		.050	5.59	4.74	4.35	4.12	3.97	3.87	3.79	3.73
		.025	8.07	6.54	5.89	5.52	5.29	5.12	4.99	4.90
		.010	12.25	9.55	8.45	7.85	7.46	7.19	6.99	6.84
		.001	29.25	21.69	18.77	17.20	16.21	15.52	15.02	14.63

Table entry for p is the critical value F^* with probability p lying to its right.

**TABLE E****F critical values (continued)**

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
60.19	60.71	61.22	61.74	62.05	62.26	62.53	62.69	62.79	63.06	63.30
241.88	243.91	245.95	248.01	249.26	250.10	251.14	251.77	252.20	253.25	254.19
968.63	976.71	984.87	993.10	998.08	1001.4	1005.6	1008.1	1009.8	1014.0	1017.7
6055.8	6106.3	6157.3	6208.7	6239.8	6260.6	6286.8	6302.5	6313.0	6339.4	6362.7
605621	610668	615764	620908	624017	626099	628712	630285	631337	633972	636301
9.39	9.41	9.42	9.44	9.45	9.46	9.47	9.47	9.47	9.48	9.49
19.40	19.41	19.43	19.45	19.46	19.46	19.47	19.48	19.48	19.49	19.49
39.40	39.41	39.43	39.45	39.46	39.46	39.47	39.48	39.48	39.49	39.50
99.40	99.42	99.43	99.45	99.46	99.47	99.47	99.48	99.48	99.49	99.50
999.40	999.42	999.43	999.45	999.46	999.47	999.47	999.48	999.48	999.49	999.50
5.23	5.22	5.20	5.18	5.17	5.17	5.16	5.15	5.15	5.14	5.13
8.79	8.74	8.70	8.66	8.63	8.62	8.59	8.58	8.57	8.55	8.53
14.42	14.34	14.25	14.17	14.12	14.08	14.04	14.01	13.99	13.95	13.91
27.23	27.05	26.87	26.69	26.58	26.50	26.41	26.35	26.32	26.22	26.14
129.25	128.32	127.37	126.42	125.84	125.45	124.96	124.66	124.47	123.97	123.53
3.92	3.90	3.87	3.84	3.83	3.82	3.80	3.80	3.79	3.78	3.76
5.96	5.91	5.86	5.80	5.77	5.75	5.72	5.70	5.69	5.66	5.63
8.84	8.75	8.66	8.56	8.50	8.46	8.41	8.38	8.36	8.31	8.26
14.55	14.37	14.20	14.02	13.91	13.84	13.75	13.69	13.65	13.56	13.47
48.05	47.41	46.76	46.10	45.70	45.43	45.09	44.88	44.75	44.40	44.09
3.30	3.27	3.24	3.21	3.19	3.17	3.16	3.15	3.14	3.12	3.11
4.74	4.68	4.62	4.56	4.52	4.50	4.46	4.44	4.43	4.40	4.37
6.62	6.52	6.43	6.33	6.27	6.23	6.18	6.14	6.12	6.07	6.02
10.05	9.89	9.72	9.55	9.45	9.38	9.29	9.24	9.20	9.11	9.03
26.92	26.42	25.91	25.39	25.08	24.87	24.60	24.44	24.33	24.06	23.82
2.94	2.90	2.87	2.84	2.81	2.80	2.78	2.77	2.76	2.74	2.72
4.06	4.00	3.94	3.87	3.83	3.81	3.77	3.75	3.74	3.70	3.67
5.46	5.37	5.27	5.17	5.11	5.07	5.01	4.98	4.96	4.90	4.86
7.87	7.72	7.56	7.40	7.30	7.23	7.14	7.09	7.06	6.97	6.89
18.41	17.99	17.56	17.12	16.85	16.67	16.44	16.31	16.21	15.98	15.77
2.70	2.67	2.63	2.59	2.57	2.56	2.54	2.52	2.51	2.49	2.47
3.64	3.57	3.51	3.44	3.40	3.38	3.34	3.32	3.30	3.27	3.23
4.76	4.67	4.57	4.47	4.40	4.36	4.31	4.28	4.25	4.20	4.15
6.62	6.47	6.31	6.16	6.06	5.99	5.91	5.86	5.82	5.74	5.66
14.08	13.71	13.32	12.93	12.69	12.53	12.33	12.20	12.12	11.91	11.72

(Continued)

TABLE E

F critical values (continued)

		Degrees of freedom in the numerator								
<i>p</i>		1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	8	.100	3.46	3.11	2.92	2.81	2.73	2.67	2.59	2.56
		.050	5.32	4.46	4.07	3.84	3.69	3.58	3.44	3.39
		.025	7.57	6.06	5.42	5.05	4.82	4.65	4.43	4.36
		.010	11.26	8.65	7.59	7.01	6.63	6.37	6.03	5.91
		.001	25.41	18.49	15.83	14.39	13.48	12.86	12.05	11.77
	9	.100	3.36	3.01	2.81	2.69	2.61	2.55	2.51	2.44
		.050	5.12	4.26	3.86	3.63	3.48	3.37	3.29	3.18
		.025	7.21	5.71	5.08	4.72	4.48	4.32	4.20	4.03
		.010	10.56	8.02	6.99	6.42	6.06	5.80	5.61	5.35
		.001	22.86	16.39	13.90	12.56	11.71	11.13	10.70	10.11
	10	.100	3.29	2.92	2.73	2.61	2.52	2.46	2.41	2.35
		.050	4.96	4.10	3.71	3.48	3.33	3.22	3.14	3.02
		.025	6.94	5.46	4.83	4.47	4.24	4.07	3.95	3.78
		.010	10.04	7.56	6.55	5.99	5.64	5.39	5.20	4.94
		.001	21.04	14.91	12.55	11.28	10.48	9.93	9.52	8.96
	11	.100	3.23	2.86	2.66	2.54	2.45	2.39	2.34	2.27
		.050	4.84	3.98	3.59	3.36	3.20	3.09	3.01	2.90
		.025	6.72	5.26	4.63	4.28	4.04	3.88	3.76	3.59
		.010	9.65	7.21	6.22	5.67	5.32	5.07	4.89	4.63
		.001	19.69	13.81	11.56	10.35	9.58	9.05	8.66	8.12
	12	.100	3.18	2.81	2.61	2.48	2.39	2.33	2.28	2.21
		.050	4.75	3.89	3.49	3.26	3.11	3.00	2.91	2.80
		.025	6.55	5.10	4.47	4.12	3.89	3.73	3.61	3.44
		.010	9.33	6.93	5.95	5.41	5.06	4.82	4.64	4.39
		.001	18.64	12.97	10.80	9.63	8.89	8.38	8.00	7.48
	13	.100	3.14	2.76	2.56	2.43	2.35	2.28	2.23	2.16
		.050	4.67	3.81	3.41	3.18	3.03	2.92	2.83	2.71
		.025	6.41	4.97	4.35	4.00	3.77	3.60	3.48	3.31
		.010	9.07	6.70	5.74	5.21	4.86	4.62	4.44	4.19
		.001	17.82	12.31	10.21	9.07	8.35	7.86	7.49	6.98
	14	.100	3.10	2.73	2.52	2.39	2.31	2.24	2.19	2.12
		.050	4.60	3.74	3.34	3.11	2.96	2.85	2.76	2.65
		.025	6.30	4.86	4.24	3.89	3.66	3.50	3.38	3.21
		.010	8.86	6.51	5.56	5.04	4.69	4.46	4.28	4.03
		.001	17.14	11.78	9.73	8.62	7.92	7.44	7.08	6.58
	15	.100	3.07	2.70	2.49	2.36	2.27	2.21	2.16	2.09
		.050	4.54	3.68	3.29	3.06	2.90	2.79	2.71	2.59
		.025	6.20	4.77	4.15	3.80	3.58	3.41	3.29	3.12
		.010	8.68	6.36	5.42	4.89	4.56	4.32	4.14	3.89
		.001	16.59	11.34	9.34	8.25	7.57	7.09	6.74	6.26
	16	.100	3.05	2.67	2.46	2.33	2.24	2.18	2.13	2.06
		.050	4.49	3.63	3.24	3.01	2.85	2.74	2.66	2.54
		.025	6.12	4.69	4.08	3.73	3.50	3.34	3.22	3.05
		.010	8.53	6.23	5.29	4.77	4.44	4.20	4.03	3.78
		.001	16.12	10.97	9.01	7.94	7.27	6.80	6.46	5.98
	17	.100	3.03	2.64	2.44	2.31	2.22	2.15	2.10	2.03
		.050	4.45	3.59	3.20	2.96	2.81	2.70	2.61	2.49
		.025	6.04	4.62	4.01	3.66	3.44	3.28	3.16	2.98
		.010	8.40	6.11	5.19	4.67	4.34	4.10	3.93	3.68
		.001	15.72	10.66	8.73	7.68	7.02	6.56	6.22	5.75

TABLE E**F critical values (continued)**

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
2.54	2.50	2.46	2.42	2.40	2.38	2.36	2.35	2.34	2.32	2.30
3.35	3.28	3.22	3.15	3.11	3.08	3.04	3.02	3.01	2.97	2.93
4.30	4.20	4.10	4.00	3.94	3.89	3.84	3.81	3.78	3.73	3.68
5.81	5.67	5.52	5.36	5.26	5.20	5.12	5.07	5.03	4.95	4.87
11.54	11.19	10.84	10.48	10.26	10.11	9.92	9.80	9.73	9.53	9.36
2.42	2.38	2.34	2.30	2.27	2.25	2.23	2.22	2.21	2.18	2.16
3.14	3.07	3.01	2.94	2.89	2.86	2.83	2.80	2.79	2.75	2.71
3.96	3.87	3.77	3.67	3.60	3.56	3.51	3.47	3.45	3.39	3.34
5.26	5.11	4.96	4.81	4.71	4.65	4.57	4.52	4.48	4.40	4.32
9.89	9.57	9.24	8.90	8.69	8.55	8.37	8.26	8.19	8.00	7.84
2.32	2.28	2.24	2.20	2.17	2.16	2.13	2.12	2.11	2.08	2.06
2.98	2.91	2.85	2.77	2.73	2.70	2.66	2.64	2.62	2.58	2.54
3.72	3.62	3.52	3.42	3.35	3.31	3.26	3.22	3.20	3.14	3.09
4.85	4.71	4.56	4.41	4.31	4.25	4.17	4.12	4.08	4.00	3.92
8.75	8.45	8.13	7.80	7.60	7.47	7.30	7.19	7.12	6.94	6.78
2.25	2.21	2.17	2.12	2.10	2.08	2.05	2.04	2.03	2.00	1.98
2.85	2.79	2.72	2.65	2.60	2.57	2.53	2.51	2.49	2.45	2.41
3.53	3.43	3.33	3.23	3.16	3.12	3.06	3.03	3.00	2.94	2.89
4.54	4.40	4.25	4.10	4.01	3.94	3.86	3.81	3.78	3.69	3.61
7.92	7.63	7.32	7.01	6.81	6.68	6.52	6.42	6.35	6.18	6.02
2.19	2.15	2.10	2.06	2.03	2.01	1.99	1.97	1.96	1.93	1.91
2.75	2.69	2.62	2.54	2.50	2.47	2.43	2.40	2.38	2.34	2.30
3.37	3.28	3.18	3.07	3.01	2.96	2.91	2.87	2.85	2.79	2.73
4.30	4.16	4.01	3.86	3.76	3.70	3.62	3.57	3.54	3.45	3.37
7.29	7.00	6.71	6.40	6.22	6.09	5.93	5.83	5.76	5.59	5.44
2.14	2.10	2.05	2.01	1.98	1.96	1.93	1.92	1.90	1.88	1.85
2.67	2.60	2.53	2.46	2.41	2.38	2.34	2.31	2.30	2.25	2.21
3.25	3.15	3.05	2.95	2.88	2.84	2.78	2.74	2.72	2.66	2.60
4.10	3.96	3.82	3.66	3.57	3.51	3.43	3.38	3.34	3.25	3.18
6.80	6.52	6.23	5.93	5.75	5.63	5.47	5.37	5.30	5.14	4.99
2.10	2.05	2.01	1.96	1.93	1.91	1.89	1.87	1.86	1.83	1.80
2.60	2.53	2.46	2.39	2.34	2.31	2.27	2.24	2.22	2.18	2.14
3.15	3.05	2.95	2.84	2.78	2.73	2.67	2.64	2.61	2.55	2.50
3.94	3.80	3.66	3.51	3.41	3.35	3.27	3.22	3.18	3.09	3.02
6.40	6.13	5.85	5.56	5.38	5.25	5.10	5.00	4.94	4.77	4.62
2.06	2.02	1.97	1.92	1.89	1.87	1.85	1.83	1.82	1.79	1.76
2.54	2.48	2.40	2.33	2.28	2.25	2.20	2.18	2.16	2.11	2.07
3.06	2.96	2.86	2.76	2.69	2.64	2.59	2.55	2.52	2.46	2.40
3.80	3.67	3.52	3.37	3.28	3.21	3.13	3.08	3.05	2.96	2.88
6.08	5.81	5.54	5.25	5.07	4.95	4.80	4.70	4.64	4.47	4.33
2.03	1.99	1.94	1.89	1.86	1.84	1.81	1.79	1.78	1.75	1.72
2.49	2.42	2.35	2.28	2.23	2.19	2.15	2.12	2.11	2.06	2.02
2.99	2.89	2.79	2.68	2.61	2.57	2.51	2.47	2.45	2.38	2.32
3.69	3.55	3.41	3.26	3.16	3.10	3.02	2.97	2.93	2.84	2.76
5.81	5.55	5.27	4.99	4.82	4.70	4.54	4.45	4.39	4.23	4.08
2.00	1.96	1.91	1.86	1.83	1.81	1.78	1.76	1.75	1.72	1.69
2.45	2.38	2.31	2.23	2.18	2.15	2.10	2.08	2.06	2.01	1.97
2.92	2.82	2.72	2.62	2.55	2.50	2.44	2.41	2.38	2.32	2.26
3.59	3.46	3.31	3.16	3.07	3.00	2.92	2.87	2.83	2.75	2.66
5.58	5.32	5.05	4.78	4.60	4.48	4.33	4.24	4.18	4.02	3.87

(Continued)

TABLE E

F critical values (continued)

		Degrees of freedom in the numerator								
<i>p</i>		1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	18	.100	3.01	2.62	2.42	2.29	2.20	2.13	2.08	2.04
		.050	4.41	3.55	3.16	2.93	2.77	2.66	2.58	2.51
		.025	5.98	4.56	3.95	3.61	3.38	3.22	3.10	3.01
		.010	8.29	6.01	5.09	4.58	4.25	4.01	3.84	3.71
		.001	15.38	10.39	8.49	7.46	6.81	6.35	6.02	5.76
	19	.100	2.99	2.61	2.40	2.27	2.18	2.11	2.06	2.02
		.050	4.38	3.52	3.13	2.90	2.74	2.63	2.54	2.48
		.025	5.92	4.51	3.90	3.56	3.33	3.17	3.05	2.96
		.010	8.18	5.93	5.01	4.50	4.17	3.94	3.77	3.63
		.001	15.08	10.16	8.28	7.27	6.62	6.18	5.85	5.59
	20	.100	2.97	2.59	2.38	2.25	2.16	2.09	2.04	2.00
		.050	4.35	3.49	3.10	2.87	2.71	2.60	2.51	2.45
		.025	5.87	4.46	3.86	3.51	3.29	3.13	3.01	2.91
		.010	8.10	5.85	4.94	4.43	4.10	3.87	3.70	3.56
		.001	14.82	9.95	8.10	7.10	6.46	6.02	5.69	5.44
	21	.100	2.96	2.57	2.36	2.23	2.14	2.08	2.02	1.98
		.050	4.32	3.47	3.07	2.84	2.68	2.57	2.49	2.42
		.025	5.83	4.42	3.82	3.48	3.25	3.09	2.97	2.87
		.010	8.02	5.78	4.87	4.37	4.04	3.81	3.64	3.51
		.001	14.59	9.77	7.94	6.95	6.32	5.88	5.56	5.31
	22	.100	2.95	2.56	2.35	2.22	2.13	2.06	2.01	1.97
		.050	4.30	3.44	3.05	2.82	2.66	2.55	2.46	2.40
		.025	5.79	4.38	3.78	3.44	3.22	3.05	2.93	2.84
		.010	7.95	5.72	4.82	4.31	3.99	3.76	3.59	3.45
		.001	14.38	9.61	7.80	6.81	6.19	5.76	5.44	5.19
	23	.100	2.94	2.55	2.34	2.21	2.11	2.05	1.99	1.95
		.050	4.28	3.42	3.03	2.80	2.64	2.53	2.44	2.37
		.025	5.75	4.35	3.75	3.41	3.18	3.02	2.90	2.81
		.010	7.88	5.66	4.76	4.26	3.94	3.71	3.54	3.41
		.001	14.20	9.47	7.67	6.70	6.08	5.65	5.33	5.09
	24	.100	2.93	2.54	2.33	2.19	2.10	2.04	1.98	1.94
		.050	4.26	3.40	3.01	2.78	2.62	2.51	2.42	2.36
		.025	5.72	4.32	3.72	3.38	3.15	2.99	2.87	2.78
		.010	7.82	5.61	4.72	4.22	3.90	3.67	3.50	3.36
		.001	14.03	9.34	7.55	6.59	5.98	5.55	5.23	4.99
	25	.100	2.92	2.53	2.32	2.18	2.09	2.02	1.97	1.93
		.050	4.24	3.39	2.99	2.76	2.60	2.49	2.40	2.34
		.025	5.69	4.29	3.69	3.35	3.13	2.97	2.85	2.75
		.010	7.77	5.57	4.68	4.18	3.85	3.63	3.46	3.32
		.001	13.88	9.22	7.45	6.49	5.89	5.46	5.15	4.91
	26	.100	2.91	2.52	2.31	2.17	2.08	2.01	1.96	1.92
		.050	4.23	3.37	2.98	2.74	2.59	2.47	2.39	2.32
		.025	5.66	4.27	3.67	3.33	3.10	2.94	2.82	2.73
		.010	7.72	5.53	4.64	4.14	3.82	3.59	3.42	3.29
		.001	13.74	9.12	7.36	6.41	5.80	5.38	5.07	4.83
	27	.100	2.90	2.51	2.30	2.17	2.07	2.00	1.95	1.91
		.050	4.21	3.35	2.96	2.73	2.57	2.46	2.37	2.31
		.025	5.63	4.24	3.65	3.31	3.08	2.92	2.80	2.71
		.010	7.68	5.49	4.60	4.11	3.78	3.56	3.39	3.26
		.001	13.61	9.02	7.27	6.33	5.73	5.31	5.00	4.76

TABLE E**F critical values (continued)**

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
1.98	1.93	1.89	1.84	1.80	1.78	1.75	1.74	1.72	1.69	1.66
2.41	2.34	2.27	2.19	2.14	2.11	2.06	2.04	2.02	1.97	1.92
2.87	2.77	2.67	2.56	2.49	2.44	2.38	2.35	2.32	2.26	2.20
3.51	3.37	3.23	3.08	2.98	2.92	2.84	2.78	2.75	2.66	2.58
5.39	5.13	4.87	4.59	4.42	4.30	4.15	4.06	4.00	3.84	3.69
1.96	1.91	1.86	1.81	1.78	1.76	1.73	1.71	1.70	1.67	1.64
2.38	2.31	2.23	2.16	2.11	2.07	2.03	2.00	1.98	1.93	1.88
2.82	2.72	2.62	2.51	2.44	2.39	2.33	2.30	2.27	2.20	2.14
3.43	3.30	3.15	3.00	2.91	2.84	2.76	2.71	2.67	2.58	2.50
5.22	4.97	4.70	4.43	4.26	4.14	3.99	3.90	3.84	3.68	3.53
1.94	1.89	1.84	1.79	1.76	1.74	1.71	1.69	1.68	1.64	1.61
2.35	2.28	2.20	2.12	2.07	2.04	1.99	1.97	1.95	1.90	1.85
2.77	2.68	2.57	2.46	2.40	2.35	2.29	2.25	2.22	2.16	2.09
3.37	3.23	3.09	2.94	2.84	2.78	2.69	2.64	2.61	2.52	2.43
5.08	4.82	4.56	4.29	4.12	4.00	3.86	3.77	3.70	3.54	3.40
1.92	1.87	1.83	1.78	1.74	1.72	1.69	1.67	1.66	1.62	1.59
2.32	2.25	2.18	2.10	2.05	2.01	1.96	1.94	1.92	1.87	1.82
2.73	2.64	2.53	2.42	2.36	2.31	2.25	2.21	2.18	2.11	2.05
3.31	3.17	3.03	2.88	2.79	2.72	2.64	2.58	2.55	2.46	2.37
4.95	4.70	4.44	4.17	4.00	3.88	3.74	3.64	3.58	3.42	3.28
1.90	1.86	1.81	1.76	1.73	1.70	1.67	1.65	1.64	1.60	1.57
2.30	2.23	2.15	2.07	2.02	1.98	1.94	1.91	1.89	1.84	1.79
2.70	2.60	2.50	2.39	2.32	2.27	2.21	2.17	2.14	2.08	2.01
3.26	3.12	2.98	2.83	2.73	2.67	2.58	2.53	2.50	2.40	2.32
4.83	4.58	4.33	4.06	3.89	3.78	3.63	3.54	3.48	3.32	3.17
1.89	1.84	1.80	1.74	1.71	1.69	1.66	1.64	1.62	1.59	1.55
2.27	2.20	2.13	2.05	2.00	1.96	1.91	1.88	1.86	1.81	1.76
2.67	2.57	2.47	2.36	2.29	2.24	2.18	2.14	2.11	2.04	1.98
3.21	3.07	2.93	2.78	2.69	2.62	2.54	2.48	2.45	2.35	2.27
4.73	4.48	4.23	3.96	3.79	3.68	3.53	3.44	3.38	3.22	3.08
1.88	1.83	1.78	1.73	1.70	1.67	1.64	1.62	1.61	1.57	1.54
2.25	2.18	2.11	2.03	1.97	1.94	1.89	1.86	1.84	1.79	1.74
2.64	2.54	2.44	2.33	2.26	2.21	2.15	2.11	2.08	2.01	1.94
3.17	3.03	2.89	2.74	2.64	2.58	2.49	2.44	2.40	2.31	2.22
4.64	4.39	4.14	3.87	3.71	3.59	3.45	3.36	3.29	3.14	2.99
1.87	1.82	1.77	1.72	1.68	1.66	1.63	1.61	1.59	1.56	1.52
2.24	2.16	2.09	2.01	1.96	1.92	1.87	1.84	1.82	1.77	1.72
2.61	2.51	2.41	2.30	2.23	2.18	2.12	2.08	2.05	1.98	1.91
3.13	2.99	2.85	2.70	2.60	2.54	2.45	2.40	2.36	2.27	2.18
4.56	4.31	4.06	3.79	3.63	3.52	3.37	3.28	3.22	3.06	2.91
1.86	1.81	1.76	1.71	1.67	1.65	1.61	1.59	1.58	1.54	1.51
2.22	2.15	2.07	1.99	1.94	1.90	1.85	1.82	1.80	1.75	1.70
2.59	2.49	2.39	2.28	2.21	2.16	2.09	2.05	2.03	1.95	1.89
3.09	2.96	2.81	2.66	2.57	2.50	2.42	2.36	2.33	2.23	2.14
4.48	4.24	3.99	3.72	3.56	3.44	3.30	3.21	3.15	2.99	2.84
1.85	1.80	1.75	1.70	1.66	1.64	1.60	1.58	1.57	1.53	1.50
2.20	2.13	2.06	1.97	1.92	1.88	1.84	1.81	1.79	1.73	1.68
2.57	2.47	2.36	2.25	2.18	2.13	2.07	2.03	2.00	1.93	1.86
3.06	2.93	2.78	2.63	2.54	2.47	2.38	2.33	2.29	2.20	2.11
4.41	4.17	3.92	3.66	3.49	3.38	3.23	3.14	3.08	2.92	2.78

(Continued)

TABLE E
F critical values (continued)

		Degrees of freedom in the numerator								
<i>p</i>		1	2	3	4	5	6	7	8	9
Degrees of freedom in the denominator	28	.100 4.20 5.61 7.64 13.50	2.50 3.34 4.22 5.45 8.93	2.29 2.95 3.63 4.57 7.19	2.16 2.71 3.29 4.07 6.25	2.06 2.56 3.06 3.75 5.66	2.00 2.45 2.90 3.53 5.24	1.94 2.36 2.78 3.36 4.93	1.90 2.29 2.69 3.23 4.69	1.87 2.24 2.61 3.12 4.50
	29	.100 4.18 5.59 7.60 13.39	2.50 3.33 4.20 5.42 8.85	2.28 2.93 3.61 4.54 7.12	2.15 2.70 3.27 4.04 6.19	2.06 2.55 3.04 3.73 5.59	1.99 2.43 2.88 3.50 5.18	1.93 2.35 2.76 3.33 4.87	1.89 2.28 2.67 3.20 4.64	1.86 2.22 2.59 3.09 4.45
	30	.100 4.17 5.57 7.56 13.29	2.49 3.32 4.18 5.39 8.77	2.28 2.92 3.59 4.51 7.05	2.14 2.69 3.25 4.02 6.12	2.05 2.53 3.03 3.70 5.53	1.98 2.42 2.87 3.47 5.12	1.93 2.33 2.75 3.30 4.82	1.88 2.27 2.65 3.17 4.58	1.85 2.21 2.57 3.07 4.39
	40	.100 4.08 5.42 7.31 12.61	2.44 3.23 4.05 5.18 8.25	2.23 2.84 3.46 4.31 6.59	2.09 2.61 3.13 3.83 5.70	2.00 2.45 2.90 3.51 5.13	1.93 2.34 2.74 3.29 4.73	1.87 2.25 2.62 3.12 4.44	1.83 2.18 2.53 2.99 4.21	1.79 2.12 2.45 2.89 4.02
	50	.100 4.03 5.34 7.17 12.22	2.41 3.18 3.97 5.06 7.96	2.20 2.79 3.39 4.20 6.34	2.06 2.56 3.05 3.72 5.46	1.97 2.40 2.83 3.41 4.90	1.90 2.29 2.67 3.19 4.51	1.84 2.20 2.55 3.02 4.22	1.80 2.13 2.46 2.89 4.00	1.76 2.07 2.38 2.78 3.82
	60	.100 4.00 5.29 7.08 11.97	2.39 3.15 3.93 4.98 7.77	2.18 2.76 3.34 4.13 6.17	2.04 2.53 3.01 3.65 5.31	1.95 2.37 2.79 3.34 4.76	1.87 2.25 2.63 3.12 4.37	1.82 2.17 2.51 2.95 4.09	1.77 2.10 2.41 2.82 3.86	1.74 2.04 2.33 2.72 3.69
	100	.100 3.94 5.18 6.90 11.50	2.36 3.09 3.83 4.82 7.41	2.14 2.70 3.25 3.98 5.86	2.00 2.46 2.92 3.51 5.02	1.91 2.31 2.70 3.21 4.48	1.83 2.19 2.54 2.99 4.11	1.78 2.10 2.42 2.82 3.83	1.73 2.03 2.32 2.69 3.61	1.69 1.97 2.24 2.59 3.44
	200	.100 3.89 5.10 6.76 11.15	2.33 3.04 3.76 4.71 7.15	2.11 2.65 3.18 3.88 5.63	1.97 2.42 2.85 3.41 4.81	1.88 2.26 2.63 3.11 4.29	1.80 2.14 2.47 2.89 3.92	1.75 2.06 2.35 2.73 3.65	1.70 1.98 2.26 2.60 3.43	1.66 1.93 2.18 2.50 3.26
	1000	.100 3.85 5.04 6.66 10.89	2.31 3.00 3.70 4.63 6.96	2.09 2.61 3.13 3.80 5.46	1.95 2.38 2.80 3.34 4.65	1.85 2.22 2.58 3.04 4.14	1.78 2.11 2.42 2.82 3.78	1.72 2.02 2.30 2.66 3.51	1.68 1.95 2.20 2.53 3.30	1.64 1.89 2.13 2.43 3.13

TABLE E**F critical values (continued)**

Degrees of freedom in the numerator										
10	12	15	20	25	30	40	50	60	120	1000
1.84	1.79	1.74	1.69	1.65	1.63	1.59	1.57	1.56	1.52	1.48
2.19	2.12	2.04	1.96	1.91	1.87	1.82	1.79	1.77	1.71	1.66
2.55	2.45	2.34	2.23	2.16	2.11	2.05	2.01	1.98	1.91	1.84
3.03	2.90	2.75	2.60	2.51	2.44	2.35	2.30	2.26	2.17	2.08
4.35	4.11	3.86	3.60	3.43	3.32	3.18	3.09	3.02	2.86	2.72
1.83	1.78	1.73	1.68	1.64	1.62	1.58	1.56	1.55	1.51	1.47
2.18	2.10	2.03	1.94	1.89	1.85	1.81	1.77	1.75	1.70	1.65
2.53	2.43	2.32	2.21	2.14	2.09	2.03	1.99	1.96	1.89	1.82
3.00	2.87	2.73	2.57	2.48	2.41	2.33	2.27	2.23	2.14	2.05
4.29	4.05	3.80	3.54	3.38	3.27	3.12	3.03	2.97	2.81	2.66
1.82	1.77	1.72	1.67	1.63	1.61	1.57	1.55	1.54	1.50	1.46
2.16	2.09	2.01	1.93	1.88	1.84	1.79	1.76	1.74	1.68	1.63
2.51	2.41	2.31	2.20	2.12	2.07	2.01	1.97	1.94	1.87	1.80
2.98	2.84	2.70	2.55	2.45	2.39	2.30	2.25	2.21	2.11	2.02
4.24	4.00	3.75	3.49	3.33	3.22	3.07	2.98	2.92	2.76	2.61
1.76	1.71	1.66	1.61	1.57	1.54	1.51	1.48	1.47	1.42	1.38
2.08	2.00	1.92	1.84	1.78	1.74	1.69	1.66	1.64	1.58	1.52
2.39	2.29	2.18	2.07	1.99	1.94	1.88	1.83	1.80	1.72	1.65
2.80	2.66	2.52	2.37	2.27	2.20	2.11	2.06	2.02	1.92	1.82
3.87	3.64	3.40	3.14	2.98	2.87	2.73	2.64	2.57	2.41	2.25
1.73	1.68	1.63	1.57	1.53	1.50	1.46	1.44	1.42	1.38	1.33
2.03	1.95	1.87	1.78	1.73	1.69	1.63	1.60	1.58	1.51	1.45
2.32	2.22	2.11	1.99	1.92	1.87	1.80	1.75	1.72	1.64	1.56
2.70	2.56	2.42	2.27	2.17	2.10	2.01	1.95	1.91	1.80	1.70
3.67	3.44	3.20	2.95	2.79	2.68	2.53	2.44	2.38	2.21	2.05
1.71	1.66	1.60	1.54	1.50	1.48	1.44	1.41	1.40	1.35	1.30
1.99	1.92	1.84	1.75	1.69	1.65	1.59	1.56	1.53	1.47	1.40
2.27	2.17	2.06	1.94	1.87	1.82	1.74	1.70	1.67	1.58	1.49
2.63	2.50	2.35	2.20	2.10	2.03	1.94	1.88	1.84	1.73	1.62
3.54	3.32	3.08	2.83	2.67	2.55	2.41	2.32	2.25	2.08	1.92
1.66	1.61	1.56	1.49	1.45	1.42	1.38	1.35	1.34	1.28	1.22
1.93	1.85	1.77	1.68	1.62	1.57	1.52	1.48	1.45	1.38	1.30
2.18	2.08	1.97	1.85	1.77	1.71	1.64	1.59	1.56	1.46	1.36
2.50	2.37	2.22	2.07	1.97	1.89	1.80	1.74	1.69	1.57	1.45
3.30	3.07	2.84	2.59	2.43	2.32	2.17	2.08	2.01	1.83	1.64
1.63	1.58	1.52	1.46	1.41	1.38	1.34	1.31	1.29	1.23	1.16
1.88	1.80	1.72	1.62	1.56	1.52	1.46	1.41	1.39	1.30	1.21
2.11	2.01	1.90	1.78	1.70	1.64	1.56	1.51	1.47	1.37	1.25
2.41	2.27	2.13	1.97	1.87	1.79	1.69	1.63	1.58	1.45	1.30
3.12	2.90	2.67	2.42	2.26	2.15	2.00	1.90	1.83	1.64	1.43
1.61	1.55	1.49	1.43	1.38	1.35	1.30	1.27	1.25	1.18	1.08
1.84	1.76	1.68	1.58	1.52	1.47	1.41	1.36	1.33	1.24	1.11
2.06	1.96	1.85	1.72	1.64	1.58	1.50	1.45	1.41	1.29	1.13
2.34	2.20	2.06	1.90	1.79	1.72	1.61	1.54	1.50	1.35	1.16
2.99	2.77	2.54	2.30	2.14	2.02	1.87	1.77	1.69	1.49	1.22

Exercise 9.1 *F – Distribution*

1. Media Matrix and Jupiter Communications gathered data on the time adults and the time teens spend online during a month. The study concluded that on average, adults spend more time online than teens. Assume that a follow-up study sampled 26 adults and 30 teens. The standard deviations of the time online during a month were 94 minutes and 58 minutes, respectively. Do the sample results support the conclusion that adults have a greater variance in online time than teens? [$F_{0.05} = 1.90$]
2. A company claims that its medicine, brand A, provides faster relief from pain than another company's medicine, brand B. A random sample from each brand gave the following times (in minutes) for relief. Test at $\alpha = 0.01$ whether we can regard the variances of the populations from which samples are drawn as equal ?

Brand A	47	51	45	53	41	55	50	46
Brand B	44	48	42	45	44	42	49	

3. Pumpkins were grown under two experimental conditions. Two random samples of 11 and 9 pumpkins show the sample standard deviations of their weights as 0.8 and 0.5 respectively. Assuming that the weight distributions are normal, test the hypothesis that the true variances are equal, against the alternative that they are not at 10% level.
4. The nicotine contents in two random samples of tobacco are given below:

Sample 1	20	23	25	28	29	
Sample 2	22	27	28	30	31	36

Can we regard the variances of the nicotine contents of the populations from which samples are drawn as equal?[LOS 5%]

5. The nicotine contents in two random samples of tobacco are given below-

Sample 1	21	24	25	26	27	
Sample 2	22	27	28	30	31	36

Can we say that the two samples came from the same population? Test at 5% LOS.

6. The following data represent a random sample of end-of-year bonuses for lower-level managerial personnel employed by a large firm. Bonuses are expressed in percentage of yearly salary.

Female	6.2	9.2	8.0	7.7	8.4	9.1	7.4	6.7
Male	8.9	10.0	9.4	8.8	12.0	9.9	11.7	9.8

Test at $\alpha = 0.01$ whether we can regard the variances of the populations from which samples are drawn as equal ?

7. Two random samples gave the following data:

Sample No.	Size	Mean	Variance
1	8	9.6	1.2
2	11	16.5	2.5

Can we conclude that the two samples have been drawn from the same normal population?[LOS 5%]

Answers

1. The adults have a greater variance in online time than teens.
2. The variances of the populations from which samples are drawn as equal.
3. The true variances are equal.
4. The variances of the populations from which samples are drawn as equal.
5. The two samples came from the same population.
6. The variances of the populations from which samples are drawn as equal.
7. The two samples have not been drawn from the same normal population.

Exercise 9.2 *One Way ANOVA*

1. Present your conclusions after doing analysis of variance to the following results of the experiment conducted in respect to five fertilizers used on plots of different fertility. [$F_{0.05} = 2.78$]

A	B	C	D	E
16	10	16	9	9
E	C	A	B	D
10	9	14	12	11
B	D	E	C	A
15	8	8	10	18
D	E	B	A	C
12	6	13	13	12
C	A	D	E	B
13	11	10	7	14

2. To test whether the mean time needed to mix a batch of material is the same for machines produced by three manufacturers, the Jacobs Chemical Company obtained the following data on the time (in minutes) needed to mix the material.

I	20	26	24	22
II	28	26	31	27
III	20	19	23	22

Use these data to test whether the population mean times for mixing a batch of material differ for the three manufacturers. [$F_{0.05} = 4.26$]

3. To study the effect of temperature on yield in a chemical process, five batches were produced at each of three temperature levels. The results follow-

50°C	60°C	70°C
34	30	23
24	31	28
36	34	28
39	23	30
32	27	31

Perform the analysis of variance on this data. Use a .05 level of significance to test whether the temperature level has an effect on the mean yield of the process. [$F_{0.05} = 3.89$]

4. The following data are from an experimental designed to investigate the perception of corporate ethical values among individuals specializing in marketing (higher scores indicate higher ethical values). Use a .05 level of significance to test for differences in perception among the three groups. Set up ANOVA table and Interpret your results. [$F_{0.05} = 3.68$]

Marketing Managers	6	5	4	5	6	4
Marketing Research	5	5	4	4	5	4
Advertising	6	7	6	5	6	6

5. To test the significance of the variation of the retail prices of a commodity in three principle cities: Mumbai, Kolkata and Delhi, four shops were chosen at random in each city and prices observed in rupees were as follows:

Mumbai	16	8	12	14
Culcutta	14	10	10	6
Delhi	4	10	8	8

Do the data indicate that the prices in the three cities are significantly different? ($F_{0.05} = 4.26$).

6. Listed below are measured amounts of greenhouse gas emissions from cars in three different categories. The measurements are in tons per year, expressed as CO_2 equivalents. Use a 0.05 significance level to test the claim that the different car categories have the same mean amount of greenhouse gas emissions. Based on the results, does the number of cylinders appear to affect the amount of greenhouse gas emissions?

Four cylinder	7.2	7.9	6.8	7.4	6.5	6.6	6.7	6.5	6.5
Six cylinder	8.7	7.7	7.7	8.7	8.2	9.0	9.3	7.4	
Eight cylinder	9.3	9.1	9.4	8.6	8.7	9.3	9.3		

7. Four salesmen A, B, C and D were posted in different areas by a company. The number of units of commodity X sold by them are as follows-

A	20	23	28	29
B	25	32	30	21
C	23	28	35	18
D	15	21	19	25

on the basis of the information, can it be concluded that there is a significant difference in the performance of the four salesmen?

8. Suppose the National Transportation Safety Board (NTSB) wants to examine the safety of compact cars, midsize cars, and full-size cars. It collects a sample of three for each of the treatments (cars types). Test whether the cars differs in pressure applied to the driver's head during a crash test. [$\alpha = 0.05$]

Compact car	Midsize car	Full size car
643	469	484
655	427	456
702	525	402
625	465	430

9. A manufacturing company has purchased three new machines of different makes and wishes to determine whether one of them is faster than the other in producing a certain output. Five hourly producing figures are observed at random from each machine and the results are given below-

Observation	M_1	M_2	M_3
1	25	31	24
2	30	39	30
3	36	38	28
4	38	42	25
5	31	35	28

Determine whether the machines are significantly different in their speed?

10. The following table gives the yields of wheat from 16 plots, all of approximately equal fertility, when 4 variety of wheat were cultivated in a completely randomized fashion. test

the hypothesis that the varieties are not significantly different.

Plot No	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Variety	A	B	D	C	B	C	A	D	B	D	A	C	B	A	B	C
Yield	32	34	29	31	33	34	34	26	36	30	33	35	37	35	35	32

11. The number of automobiles arriving at 4 toll gates were recorded for a 2 hours time period (10 : 00 – 12 : 00) for each of the six working days. The data are as follows-

	Mon	Tue	Wed	Thu	Fri	Sat
Gate 1	20	21	23	22	19	16
Gate 2	15	23	22	19	12	18
Gate 3	13	19	24	18	15	21
Gate 4	30	29	22	32	16	17

Determine whether the arrival is the same at each toll gate. Test at LOS 1%.

Answers

1. We may conclude that at least two fertilizers are differ significantly.
2. Differ significantly for the three manufacturers.
3. There is no significant difference in the mean yield of the process due to the temperature level.
4. The null hypothesis is rejected.
5. The differences between the retail prices of the commodity in the three cities are not significant.
6. The number of cylinders appear affects the amount of greenhouse gas emissions.
7. There is no significant difference in the performance of the four salesmen.
8. The cars differ in pressure applied to the driver's head during a crash test.
9. There is a significant difference in the speeds of the three machines.
10. There is a significant difference difference between the varieties.
11. The arrival is the same at each toll gate.

Exercise 9.3 *Two way ANOVA*

1. The following data represents the number of units of production per day turned out by 5 different workers using 4 different types of machines:

		Machine type			
		A	B	C	D
Workers	1	44	38	47	36
	2	46	40	52	43
	3	34	36	44	32
	4	43	38	46	33
	5	38	42	49	39

Test whether the (a) five workers differ with respect to mean productivity. (b) mean productivity is the same for four different machine types.

2. Four different drugs have been developed for a certain disease. These drugs are used in 3 different hospitals and the results given below show the number of cases of recovery from the disease per 100 people who have taken drugs. Determine whether (a) the drugs are same in their effect. (b) the recovery cases differ in hospitals. [Test at 1% LOS]

Hospitals	Drugs			
	D_1	D_2	D_3	D_4
H_1	19	8	23	8
H_2	10	9	12	6
H_3	11	13	13	10

3. The following table gives the gains in weights of 4 different types of cows fed on 3 different rations over a period. Test at 5% whether (i) the difference in the rations is significant, (ii) the 4 types of cows differ significantly in gaining weight.

Ration	Types of cow			
	I	II	III	IV
A	13.8	15.7	16.0	20.2
B	8.7	11.8	9.0	12.9
C	12.0	16.5	13.3	12.5

4. Five breeds of cattle B_1, B_2, B_3, B_4, B_5 were fed on four different rations R_1, R_2, R_3, R_4 . Gains in weight in kg over a given period were recorded and are given below-

	B_1	B_2	B_3	B_4	B_5
R_1	1.9	2.2	2.6	1.8	2.1
R_2	2.5	1.9	2.3	2.6	2.2
R_3	1.7	1.9	2.2	2.0	2.1
R_4	2.1	1.8	2.5	2.3	2.4

Is there a significant difference between (i) breeds and (ii) rations?

5. Four experiments determine the moisture content of samples of a powder, each observer

taking a sample from each of the four consignments. The assessments are given below-

		Consignement			
		<i>A</i>	<i>B</i>	<i>C</i>	<i>D</i>
Observer	<i>I</i>	9	10	9	8
	<i>II</i>	12	11	9	7
	<i>III</i>	11	10	10	9

Determine whether there is any significant difference between consignments or between observers. [$\alpha = 0.05$]

Answers

- (i) the mean productivity is not the same for four different machine types.
(ii) the five workers differ with respect to mean productivity.
- (i) the drugs are same in their effect. (ii) the recovery cases are the same in hospitals.
- (i) the difference in the rations is not significant (ii) the 4 types of cows differ significantly in gaining weight.
- (i) the difference in the breeds is not significant (ii) the difference in the rations is not significant.
- (i) there is no significant difference between consignments. (ii) there is no significant difference between observers.